



TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY
BHAKTAPUR MULTIPLE CAMPUS

A Project Report on
“Cybercrime Reporting System”

In partial fulfillment of the requirements for the degree of
Bachelor of Information Technology (BIT)

Under the Supervision of

Mr. Surya Bam

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by Mr. Sakshyam Karki, Mr. Bikram Gharti, and Mr. Tanish Upadhyay, entitled **Cybercrime Reporting System** in partial fulfillment of the requirement for the degree of Bachelor in Information Technology (BIT), be processed for evaluation.

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled **Cybercrime Reporting System** prepared by Mr. Sakshyam Karki (79020172), Mr. Bikram Gharti (79020151), and Mr. Tanish Upadhyay (79020179), in partial fulfillment of the requirements for the four-year Bachelor Degree in Bachelor of Information Technology, has been well studied. In our opinion, it is satisfactory in scope and quality for the required degree.

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We would also like to extend our appreciation to our team members for their hard work and dedication in completing this project.

Thank You,

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ABSTRACT

Rapid advancement in AI for the creation and modification of digital imagery have posed several new problems for cybersecurity specifically the problem of misleading visual data dissemination. Conventional methods of reporting cybercrime necessitate the manual analysis of digital evidence, which tends to be slow and inconsistent and lacks an automated way to authenticate the digital evidence itself. This research proposes a web-based application that, called "Cybercrime Reporting System", built on the Django framework and it is responsible for letting users report crimes with the optional submission of images as support for the subsequent analysis. In this work we propose an AI-based image analysis pipeline using a DenseNet121 trained on the CIFAKE dataset, responsible for classifying a submitted image in one of the three classes: Likely Manipulated, Suspicious, or Likely Real. This system's analysis also incorporates LBP for image texture analysis to provide a second dimension for interpretability before the information is used to feed a rule-based decision engine to generate a systematic AI assisted analysis report. This work embraces a human-in-the-loop system, in that the final decision will be left to a system administrator once they view the submitted report, image, and AI generated results. An overall accuracy of 98.61%, precision of 98.78%, recall of 98.45%, an F1-score of 98.61%, and an AUC-ROC of 99.79% were observed on the CIFAKE test set, showing that there is good potential to be highly reliable in fake image identification, particularly for those images generated by AI.

Keywords: Cybercrime reporting, image authenticity, deepfake detection, DenseNet121, CIFAKE dataset, Local Binary Patterns (LBP), human-in-the-loop, AI-assisted analysis

TABLE OF CONTENTS

SUPERVISOR’S RECOMMENDATION	i
CERTIFICATE OF APPROVAL	ii
ACKNOWLEDGEMENT.....	iii
ABSTRACT	iv
TABLE OF CONTENTS.....	v
LIST OF ABBREVIATIONS	vii
LIST OF FIGURES	viii
LIST OF TABLES.....	x
CHAPTER 1: INTRODUCTION.....	1
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Objectives.....	2
1.4 Scope and Limitations	2
1.4.1 Scope	2
1.4.2 Limitations.....	2
1.5 Development Methodology.....	3
1.6 Report Organization	3
CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW.....	4
2.1 Background Study	4
2.2 Literature Review	5
CHAPTER 3: SYSTEM ANALYSIS.....	8
3.1 Requirement Analysis	8
3.1.1 Functional Requirements.....	8
3.1.2 Non-Functional Requirements.....	9
3.2 Feasibility Study.....	9
3.2.1 Technical Feasibility	9
3.2.2 Operational Feasibility	10
3.2.3 Economic Feasibility	10
3.2.4 Schedule Feasibility.....	10
3.3 Analysis	10
3.3.1 CIFAKE Dataset	10
3.3.2 Data Modelling: Entity-Relationship Diagram.....	12

3.3.3 Process Modelling: Data Flow Diagram (DFD).....	12
CHAPTER 4: SYSTEM DESIGN	18
4.1 Architectural Design.....	18
4.2 Database Schema Design	18
4.3 Process Design	19
4.4 Forms and Report Design.....	21
4.5 Interface and Dialogue Design.....	24
4.6 Algorithm Details	27
4.6.1 DenseNet121	27
4.6.2 Local Binary Pattern (LBP).....	29
4.7 Evaluation Metrics	30
CHAPTER 5: IMPLEMENTATION AND TESTING.....	32
5.1 Implementation.....	32
5.1.1 Tools Used	32
5.1.2 Implementation Details of Modules	32
5.2 Testing	40
5.2.1 Unit Testing	40
5.2.2 System Testing.....	46
5.3 Result Analysis.....	47
CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATIONS	50
6.1 Conclusion.....	50
6.2 Future Recommendations.....	50
REFERENCES.....	52
APPENDICES	53
Log of Visits to Supervisor	56

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CIFAKE	CIFAR-10 Fake Image Dataset
CNN	Convolutional Neural Network
DFD	Data Flow Diagram
ER	Entity-Relationship
GAN	Generative Adversarial Network
JSON	JavaScript Object Notation
JWT	JSON Web Token
LBP	Local Binary Patterns
REST	Representational State Transfer

LIST OF FIGURES

Figure 1.1: Waterfall Model.....	3
Figure 3.1: Use Case Diagram.....	9
Figure 3.2: Gantt Chart.....	10
Figure 3.3: Entity Relationship Diagram.....	12
Figure 3.4: DFD Level 0.....	13
Figure 3.5: DFD level 1.....	13
Figure 3.6: DFD level 2 for process 1.0.....	14
Figure 3.7: DFD level 2 for process 2.0.....	15
Figure 3.8: DFD level 2 for process 3.0.....	16
Figure 3.9: DFD level 2 for process 4.0.....	17
Figure 3.10: DFD level 2 for process 5.0.....	17
Figure 4.1 Architectural Design.....	18
Figure 4.2: Database Schema Diagram.....	19
Figure 4.3: System Flowchart.....	20
Figure 4.4: Registration Form Design.....	21
Figure 4.5: Login Form Design.....	22
Figure 4.6: Complaint Submission Form.....	22
Figure 4.7: Admin Review Form.....	23
Figure 4.8: Cybercrime Complaint Review Report.....	24
Figure 4.9: User Dashboard.....	25
Figure 4.10: Complaint Tracking Page.....	25
Figure 4.11: Admin Dashboard.....	26
Figure 4.12: Complaint Management Page.....	26
Figure 5.1: Authentication Module.....	33
Figure 5.2: Complaint submission form part 1.....	34
Figure 5.3: Complaint submission form part 2.....	34
Figure 5.4: Complaint submission form part 3.....	35
Figure 5.5: Complaint creation.....	35
Figure 5.6: AI Analysis Output from Decision Engine.....	38
Figure 5.7: Report Generation Format part 1.....	39
Figure 5.8: Report Generation Format part 2.....	39
Figure 5.9: Registration.....	41

Figure 5.10: Failed Registration.....	41
Figure 5.11: Duplicate account registration failed.....	42
Figure 5.12: Login.....	42
Figure 5.13: Successful Login popup.....	42
Figure 5.14: Failed Login.....	43
Figure 5.15: Form Submission Error Popup when input is GIF.....	44
Figure 5.16: Form submission without image upload part 1.....	44
Figure 5.17: Form submission without image upload part 2.....	45
Figure 5.18: Form submission without image upload part 3.....	45
Figure 5.19: Success form submission without image upload.....	45
Figure 5.20: Testing with real picture.....	46
Figure 5.21: Testing with low quality and mixed textures.....	46
Figure 5.22: Training and Validation Curve.....	48
Figure 5.23: Confusion Matrix.....	48

LIST OF TABLES

Table 3.1: CIFAKE Dataset Summary.....	11
Table 4.1: Densenet121 Architecture.....	27
Table 5.1: Decision Rules.....	37
Table 5.2: Authentication Testing.....	40
Table 5.3: Complaint Submission Testing.....	43
Table 5.4: AI Detection Module Testing.....	46
Table 5.5: System Testing.....	47
Table 5.6: Training Configuration.....	47
Table 5.7: Model Performance.....	48

CHAPTER 1: INTRODUCTION

1.1 Introduction

With the spread of the Internet and digital technology, the proliferation of cybercrime is one of the primary challenges in today's digital age. It has recently attracted growing attention due to the increasing abuse of altered and AI-generated images that can be used for fraudulent purposes, dissemination of misinformation, identity theft and other types of cybercrime. Existing Cybercrime Reporting systems only focus on gathering complaint information and usually resort to human-based investigation of the evidence. However, this may consume considerable time and inconsistency due to subjective human interpretation.

To tackle such issue, this project proposes a Cybercrime Reporting System with Django framework. The web application permits user registration, secure login, filing of cybercrime complaints and uploading image evidence to be submitted. The complainant can also track the complaint's progress. An administrator can browse complaint data, view image evidence, make comments and update the status using the complaint management framework.

One of the important functions implemented in this system is an AI based image authenticity pipeline. Using the CIFAKE image datasets and the DenseNet121 model for classification, the submitted images are judged as likely modified, doubtful or likely original. An additional image texture verification using Local Binary Pattern (LBP) is used to detect possible modifications of images by highlighting the differences in texture. Combining DenseNet121 and LBP decision using the rules produces a clear AI-assisted report on image authentication. A human-in-the-loop model is applied to allow administrators to make the final decision upon the review of complaints information, image evidence, and AI analysis results. This approach integrates artificial intelligence support to investigate a cybercrime complaint in an automated fashion, while allowing the final decision making to be performed by humans.

1.2 Problem Statement

Existing cybercrime reporting systems primarily function as structured platforms for complaint submission and management. However, they lack integrated mechanisms for analyzing digital evidence, particularly image-based content. As a result, law enforcement officers must rely on manual inspection to assess image authenticity, which is time-consuming, technically challenging, and prone to inconsistency.

With the growing prevalence of AI-generated and manipulated images, manual evaluation alone is no longer sufficient. There is a clear need for a system that can assist in the early-stage analysis of image evidence to improve efficiency and reliability in complaint processing.

1.3 Objectives

The objectives of this project are as follows:

- To develop a web-based cybercrime reporting system that enables users to submit complaints along with supporting image evidence.
- To implement an AI-assisted image authenticity detection module using DenseNet121 and Local Binary Pattern (LBP) to analyze submitted images and identify potential AI-generated or manipulated content.
- To provide a human-in-the-loop complaint review workflow that generates AI-assisted analysis reports to support administrators in making informed decisions.

1.4 Scope and Limitations

1.4.1 Scope

The system is designed to handle cybercrime complaints submitted through a web-based interface. Each complaint may include a single static image file (JPEG or PNG format) as supporting evidence. Two primary user roles are supported: users, who act as complainants submitting reports, and administrators, who serve as review authorities. When image evidence is provided, the AI analysis pipeline is triggered automatically, generating a classification result and an analysis report to assist administrators during the review process. The system is intended to improve the efficiency, consistency, and technical rigor of initial complaint evaluation by integrating AI-assisted image analysis directly into the reporting workflow.

1.4.2 Limitations

The AI image detection model is trained on the CIFAKE dataset, which comprises relatively low-resolution images of 64×64 pixels; this may limit detection accuracy when applied to high-resolution or advanced AI-generated images encountered in real-world submissions. LBP-based texture analysis functions as a supportive technique and is not capable of independently detecting manipulated or AI-generated images. The system processes only static JPEG and PNG images and does not currently support video, audio, or document-based evidence. Decision thresholds applied within the rule-based engine are empirically defined and may require adjustment to generalize across diverse image types or evolving manipulation techniques. The system is designed to function as a decision-support tool and does not produce legally binding or forensically certified determinations; final decisions remain with the reviewing administrator.

1.5 Development Methodology

This project follows the Waterfall development model, where development is carried out in sequential phases. Each phase is completed before moving to the next, ensuring a structured and systematic approach.

The phases include:

1. **Requirement Analysis:** Identification of system requirements and analysis of existing solutions.
2. **System Design:** Design of system architecture, database schema, and AI pipeline.
3. **Implementation:** Development of the web application and integration of AI modules.
4. **Testing:** Unit and system testing to validate functionality and performance.

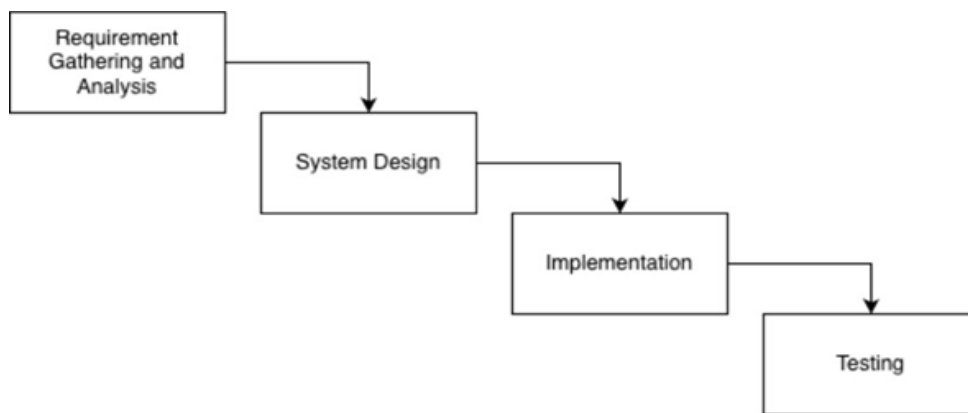


Figure 1.1: Waterfall Model

1.6 Report Organization

This report is organized into six chapters. Chapter One introduces the project by presenting the background, problem statement, objectives, scope, limitations, and significance of the proposed system. Chapter Two presents the background study and literature review by discussing existing cybercrime reporting systems, relevant technologies, previous research, and the research gaps addressed by the proposed system. Chapter Three focuses on system analysis and requirements, including the analysis of the existing system, feasibility study, and the functional and non-functional requirements.

Chapter Four describes system design, including the system architecture, database design, user interface design, and implementation methodology. Chapter Five explains the development and testing of the system, covering the technologies used, implementation of major modules, and testing performed to validate the system. Finally, Chapter Six summarizes the project's findings and achievements, evaluates the accomplishment of the objectives, and provides recommendations for future improvements and further research.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Cybercrime reporting systems are web-based platforms that allow individuals and organizations to report cyber-related incidents such as phishing attacks, online fraud, identity theft, cyber harassment, hacking, and social media crimes. These systems help law enforcement agencies collect, organize, and monitor complaints in a centralized database, making the investigation process more systematic and efficient. Traditional reporting systems mainly focus on complaint registration, evidence storage, and communication between complainants and investigators. However, they generally lack intelligent mechanisms for automatically analyzing digital evidence such as images submitted with complaints, requiring investigators to perform manual verification, which is both time-consuming and resource intensive [1].

Artificial Intelligence (AI) has become one of the most significant technologies for analyzing digital information automatically. In image processing, AI enables computers to identify visual patterns, extract meaningful features, and classify images with high accuracy. AI-based image analysis has been successfully applied in healthcare, surveillance, digital forensics, autonomous systems, and cybersecurity. These applications reduce manual effort while improving the speed and consistency of decision-making. In cybercrime investigations, AI can assist investigators by automatically examining submitted image evidence and identifying whether an image is authentic or artificially generated [2].

Among various AI techniques, Convolutional Neural Networks (CNNs) have become the standard approach for image classification and recognition tasks. CNNs automatically learn hierarchical image features through multiple convolutional layers instead of relying on manually designed image descriptors. During training, the network extracts low-level features such as edges and textures before learning more complex visual patterns in deeper layers. This automatic feature learning enables CNNs to achieve superior performance compared to conventional image processing methods in image classification, object detection, and forgery detection [3].

One of the most effective CNN architectures is DenseNet121, introduced by Huang et al. DenseNet establishes dense connections between consecutive layers, allowing each layer to receive feature information from all preceding layers. This design improves feature propagation, encourages feature reuse, minimizes the vanishing gradient problem, and reduces

the number of trainable parameters compared to many traditional CNN architectures. Because of its efficient learning capability and high classification accuracy, DenseNet121 has been selected as the primary image classification model in this project for distinguishing authentic images from AI-generated images [4].

Besides deep learning, traditional texture analysis methods remain valuable for digital image forensics. Local Binary Pattern (LBP) is a feature extraction technique that describes local image texture by comparing the intensity value of a center pixel with its surrounding neighboring pixels. The resulting binary representation effectively captures texture characteristics while requiring minimal computational resources. LBP has been successfully applied in face recognition, texture classification, and image forgery detection because manipulated images often contain subtle texture inconsistencies that are difficult to observe visually. In this project, LBP is used alongside DenseNet121 to provide complementary texture-based features for image authenticity analysis [5].

Recent developments in Generative Artificial Intelligence (Generative AI) have enabled the creation of highly realistic synthetic images through deep learning models such as Generative Adversarial Networks (GANs). GANs consist of two competing neural networks a generator that produces synthetic images and a discriminator that attempts to distinguish generated images from real ones. Continuous competition between these two networks enables the generator to create increasingly realistic images. Although GANs have beneficial applications in education, entertainment, and content generation, they also present significant cybersecurity risks by enabling image forgery, identity impersonation, misinformation, and manipulation of digital evidence [6].

Based on these developments, integrating AI-powered image authenticity detection into cybercrime reporting systems can significantly improve the complaint verification process. Instead of relying solely on manual inspection, investigators can obtain automated authenticity assessments that identify whether submitted images are likely to be genuine or AI-generated. Such intelligent assistance reduces investigation time, improves preliminary evidence screening, and supports informed decision-making while ensuring that the final judgment remains under human supervision [7].

2.2 Literature Review

The rapid advancement of artificial intelligence has significantly increased research on detecting manipulated and AI-generated images. The availability of powerful image generation models has created new challenges in cybersecurity, digital forensics, misinformation

detection, and cybercrime investigations. Consequently, researchers have proposed numerous deep learning and image processing techniques for improving image authenticity verification [8].

Deep learning architectures such as DenseNet, Xception, EfficientNet, and ResNet have demonstrated excellent performance in image classification and manipulation detection tasks. Among these architectures, DenseNet has received considerable attention because of its dense connectivity mechanism, which improves feature reuse and enables efficient learning with fewer trainable parameters. Huang et al. demonstrated that DenseNet achieves competitive classification accuracy while reducing computational complexity compared to conventional CNN architectures, making it suitable for image authenticity classification tasks [4].

Another widely adopted architecture is Xception, which replaces standard convolution operations with depthwise separable convolutions to reduce computational cost while maintaining high classification performance. Researchers have extensively evaluated Xception using benchmark datasets such as FaceForensics++, where the model successfully detects manipulated facial images generated through deepfake techniques. The study demonstrated that deep CNN architectures are highly effective for identifying subtle visual artifacts introduced during image manipulation [9].

Bird and Lotfi introduced the CIFAKE dataset, which contains both authentic and AI-generated images for evaluating deep learning models. Their experiments showed that DenseNet-based architectures consistently achieved high classification accuracy when distinguishing genuine images from synthetic images. The dataset has become an important benchmark for evaluating AI-generated image detection systems and further supports the selection of DenseNet121 for this project [10].

Traditional feature extraction methods continue to play an important role in digital image forensics despite the success of deep learning. Local Binary Pattern (LBP) remains one of the most widely used texture descriptors because of its computational simplicity and ability to capture local texture information. Although LBP alone cannot reliably detect highly sophisticated AI-generated images, previous studies have shown that combining LBP with deep learning models improves overall detection performance by incorporating complementary texture information that CNNs may overlook [5].

Although considerable research has been conducted on image forgery detection and AI-generated image classification, most existing cybercrime reporting systems primarily concentrate on complaint registration, evidence management, and investigation workflow. Automated verification of submitted image evidence is generally absent, requiring

investigators to manually examine digital evidence before initiating further investigation. This limitation increases workload and may delay the complaint verification process, particularly when handling a large number of reports [1].

The reviewed literature indicates that previous studies have largely focused on improving image classification accuracy rather than integrating AI-based image authenticity detection into practical cybercrime reporting platforms. Therefore, this project addresses this research gap by incorporating a DenseNet121-based image classification model together with Local Binary Pattern (LBP) texture analysis into a cybercrime reporting system. The proposed approach provides automated image authenticity assessment to support investigators during preliminary evidence verification while ensuring that all final investigative decisions remain under human supervision.

CHAPTER 3: SYSTEM ANALYSIS

System Analysis

The system analysis phase defines the structure and operational behaviour of the Cybercrime Reporting System. Following the Waterfall model, this phase translates project objectives into clear system requirements, feasibility considerations, and overall system design.

3.1 Requirement Analysis

3.1.1 Functional Requirements

- **User Registration and Login:** The system shall allow users to register and log in securely.
- **Complaint Submission:** The system shall allow users to submit complaints with a title and description.
- **Evidence Image Upload:** The system shall allow users to upload one image (JPEG/PNG) as supporting evidence.
- **Automatic AI Analysis:** The system shall automatically perform AI-based analysis when an image is uploaded.
- **Image Authenticity Analysis Result:** The system shall generate an AI confidence score indicating whether the image is likely manipulated, suspicious, or likely real.
- **LBP Analysis Output:** The system shall perform texture-based analysis using LBP to support AI results.
- **Decision Engine Verdict:** The system shall apply rule-based logic to produce a final classification label with explanation.
- **AI Report Generation:** The system shall generate and store an AI-assisted report including analysis results and interpretation.
- **Admin Review and Status Update:** The system shall allow an admin to review complaints and update status (Approved/Rejected).
- **Report Download:** The system shall allow users to download the final report after review.

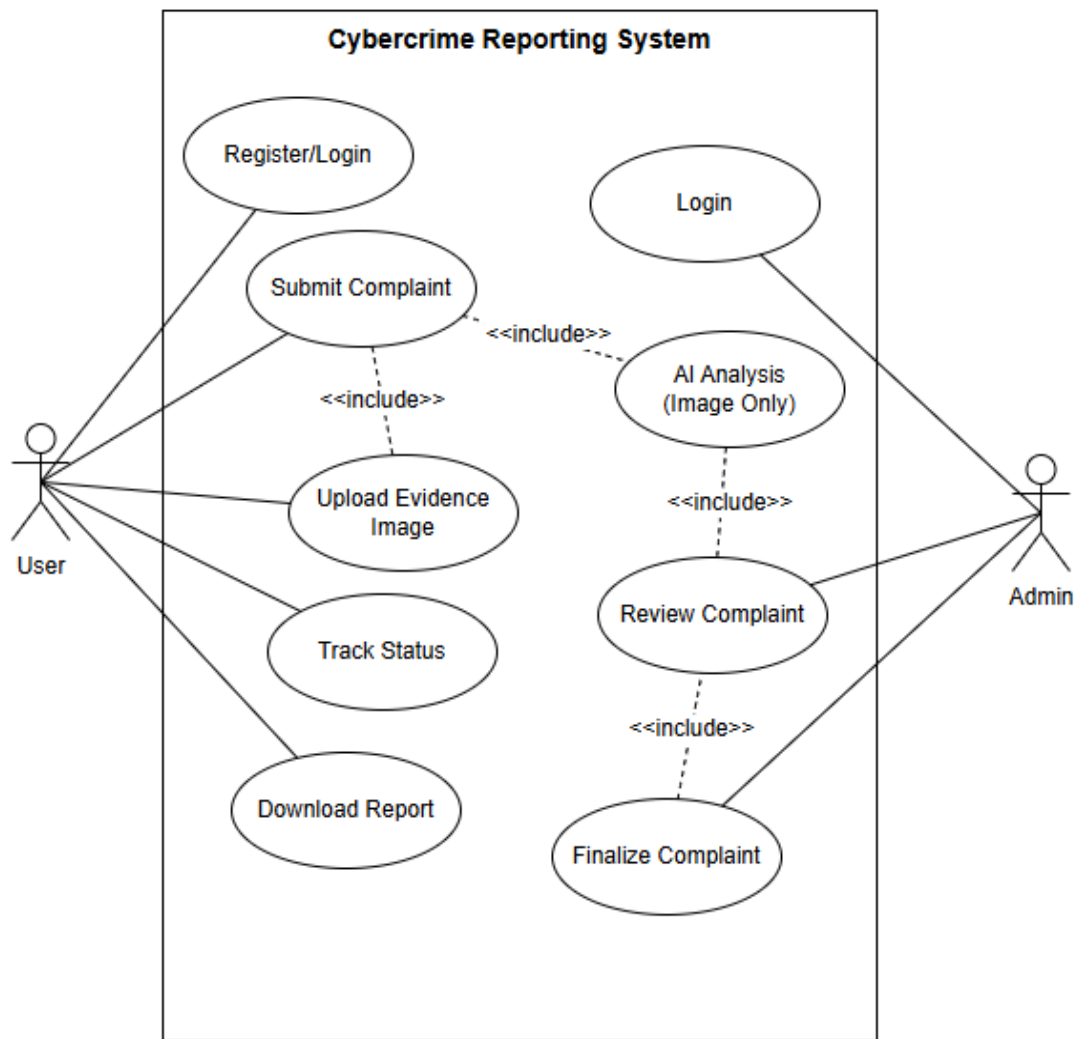


Figure 3.1: Use Case Diagram

3.1.2 Non-Functional Requirements

- **Security:**
User passwords shall be securely stored using hashing mechanisms.
- **Performance:**
AI analysis should be completed within a reasonable time.
- **Reliability:**
The system shall handle AI processing failures gracefully without data loss.
- **Usability:**
The system shall provide a simple and intuitive interface for users and administrators.

3.2 Feasibility Study

3.2.1 Technical Feasibility

The system is technically feasible using the selected technologies. Django is used for backend development, handling authentication, database operations, and complaint workflows. The AI

module is implemented using PyTorch, with DenseNet121 used for image classification based on the CIFAKE dataset. Local Binary Pattern (LBP) is implemented as a lightweight image processing technique to support texture analysis. The AI pipeline operates on resized images, enabling efficient execution on CPU within a standard development environment.

3.2.2 Operational Feasibility

The system is easy to operate for its intended users. Complainants submit complaints and optionally upload an image. The AI system automatically generates an analysis report, which is reviewed by an administrator.

The human-in-the-loop design ensures that final decisions are made by an authorized person, maintaining reliability and accountability in the complaint handling process.

3.2.3 Economic Feasibility

The project is economically feasible as it relies entirely on open-source technologies such as Django and PyTorch. The CIFAKE dataset is publicly available, and the system can be deployed using standard computing resources without additional cost.

3.2.4 Schedule Feasibility

The project is feasible within the given academic timeline as it follows a structured and organized development approach. The initial phase focuses on setting up the authentication and complaint management system, ensuring secure user access and basic system functionality. The next phase involves integrating the AI module using DenseNet121 and Local Binary Pattern (LBP) techniques for image analysis and classification. After successful integration, the project proceeds with the development of the report generation feature and the administrative panel for system management. Finally, thorough testing and documentation are carried out to verify the system's performance, reliability, and readiness for deployment.

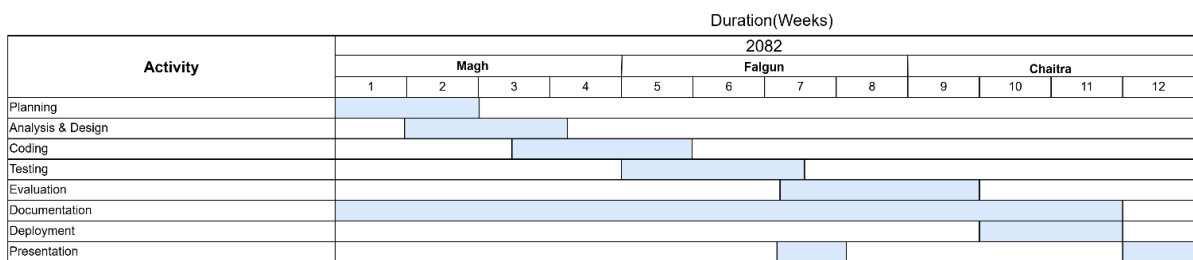


Figure 3.2: Gantt Chart

3.3 Analysis

3.3.1 CIFAKE Dataset

The DenseNet121 model was trained and evaluated using the CIFAKE dataset developed by Jordan Bird and Abdelrahman Lotfi [1]. The CIFAKE dataset is specifically designed for image

authenticity classification and contains an equal number of real and AI-generated images, making it suitable for training deep learning models to distinguish between authentic and manipulated visual content.

The dataset consists of 120,000 color images, including 60,000 real images and 60,000 AI-generated images. The real images are derived from the CIFAR-10, while the fake images are generated using Stable Diffusion. The balanced distribution of the two classes helps reduce classification bias and enables the model to learn representative features from both authentic and synthetic images.

All images in the dataset are stored in RGB color format with three color channels (Red, Green, and Blue). The original image dimensions are 32×32 pixels, but for this project, the images were resized to 64×64 pixels before being used as input to the DenseNet121 model. Resizing ensures that all images have a consistent input size, which is required for batch processing in deep learning networks.

Before training, each image underwent a preprocessing stage. The pixel values were normalized using the ImageNet mean and standard deviation values, which improves numerical stability during training and accelerates convergence. In addition, several data augmentation techniques were applied to increase the diversity of the training data and improve the model's generalization capability. These techniques included random horizontal flipping, random rotation, and color jittering.

The dataset was divided into training and testing subsets. The training set was used to train the DenseNet121 model, while the testing set was used to evaluate its performance on unseen data. The details of the dataset are summarized in Table 3.1.

Table 3.1: CIFAKE Dataset Summary

Parameter	Value
Total Images	120,000
Real Images	60,000
Fake Images	60,000
Original Image Size	32×32 pixels
Input Image Size	64×64 pixels
Color Format	RGB (3 channels)
File Format	PNG/JPEG
Normalization	ImageNet mean and standard deviation
Data Augmentation	Random horizontal flip, rotation, color jitter

Training Set	100,000 images
Testing Set	20,000 images

3.3.2 Data Modelling: Entity-Relationship Diagram

The ERD shows all seven entities, their attributes with primary/foreign key annotations, and their cardinality relationships. The User entity uses email as the login identifier and full name fields for identity no username field is present.

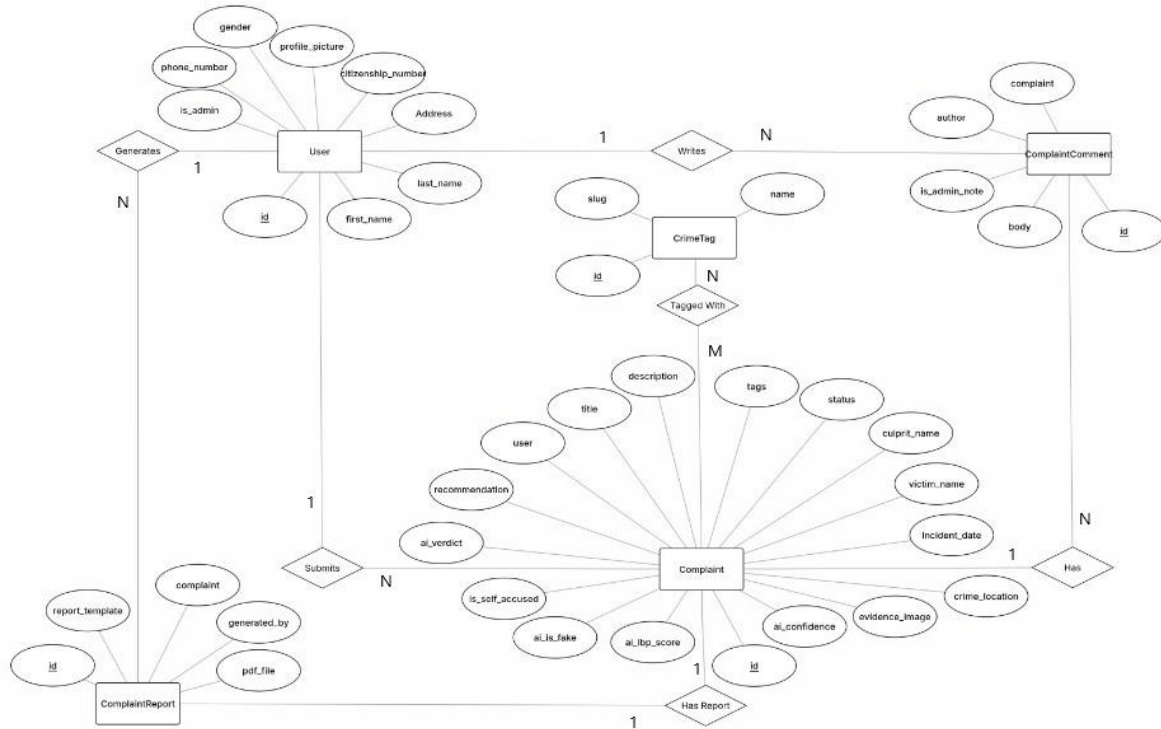


Figure 3.3: Entity Relationship Diagram

3.3.3 Process Modelling: Data Flow Diagram (DFD)

a. DFD Level 0 (Context Diagram)

The Level 0 DFD (Context Diagram) represents the entire Cybercrime Reporting System as a single process. Users interact with the system by registering, logging in, and submitting complaints with optional image evidence. The system provides complaint status updates and generates a case report for users. Administrators access complaint information and AI analysis results; review submitted cases and provide verdicts and remarks. This diagram shows the overall interaction between external entities and the system without detailing internal processes.

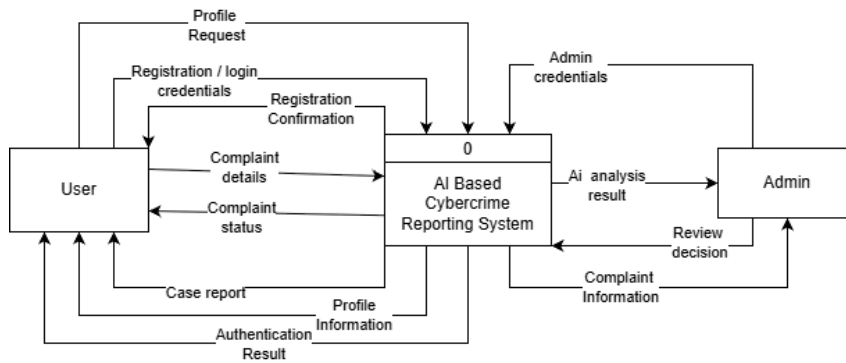


Figure 3.4: DFD Level 0

b. DFD Level 1

The Level 1 DFD decomposes the Cybercrime Reporting System into its major functional processes. User Management handles registration and authentication, while Complaint Management manages complaint submission and status tracking. When image evidence is provided, the AI Image Analysis process analyzes the image and generates AI results. Administrators review complaint details and AI findings through the Admin Review process and provide a final verdict. Finally, the Report Generation process creates a case report that can be accessed by the user. The diagram also shows the logical data stores used to manage user, complaint, AI analysis, and report information.

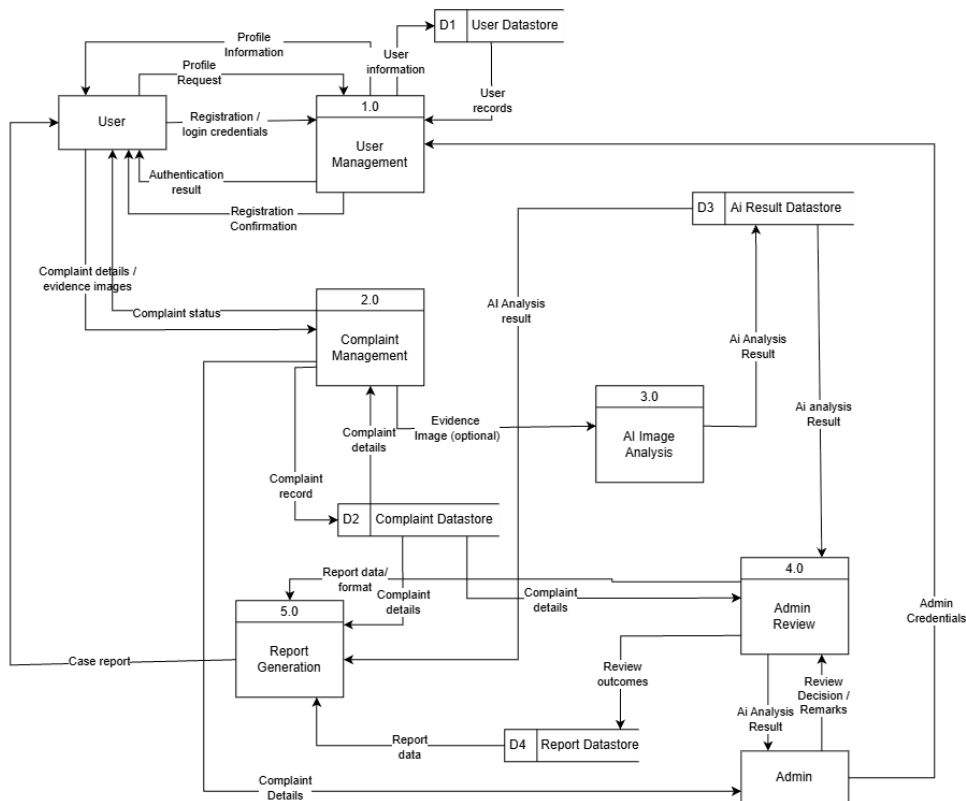


Figure 3.5: DFD level 1

c. DFD Level 2

The Level 2 DFD provides a more detailed decomposition of the Cybercrime Reporting System by illustrating the internal sub-processes of each major function. It shows how user registration and authentication, complaint submission, image evidence processing, AI image analysis, administrative review, and report generation are carried out through interconnected activities.

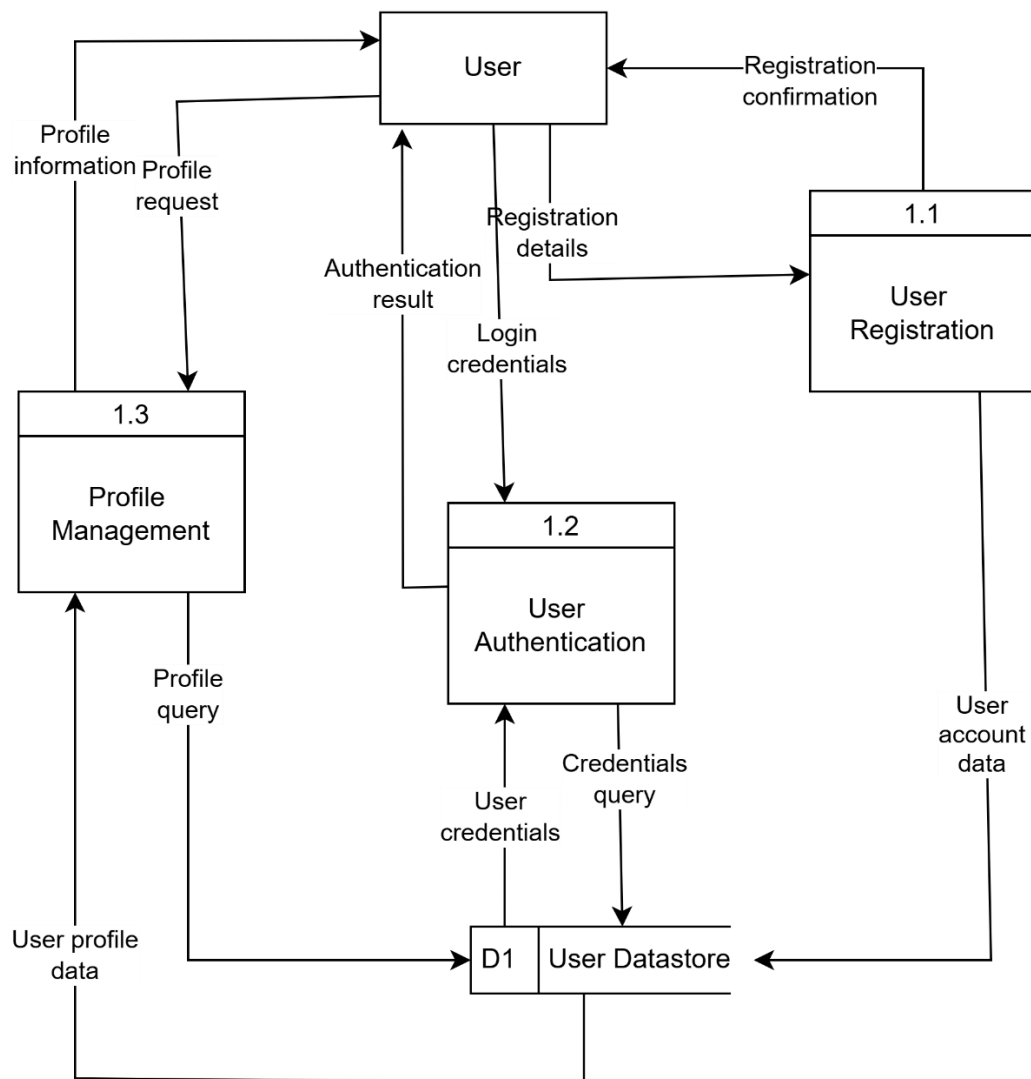


Figure 3.6: DFD level 2 for process 1.0

The Level 2 DFD of User Management illustrates the processes involved in user registration, authentication, and profile management. It manages user account information and verifies login credentials before allowing access to the system's functionalities.

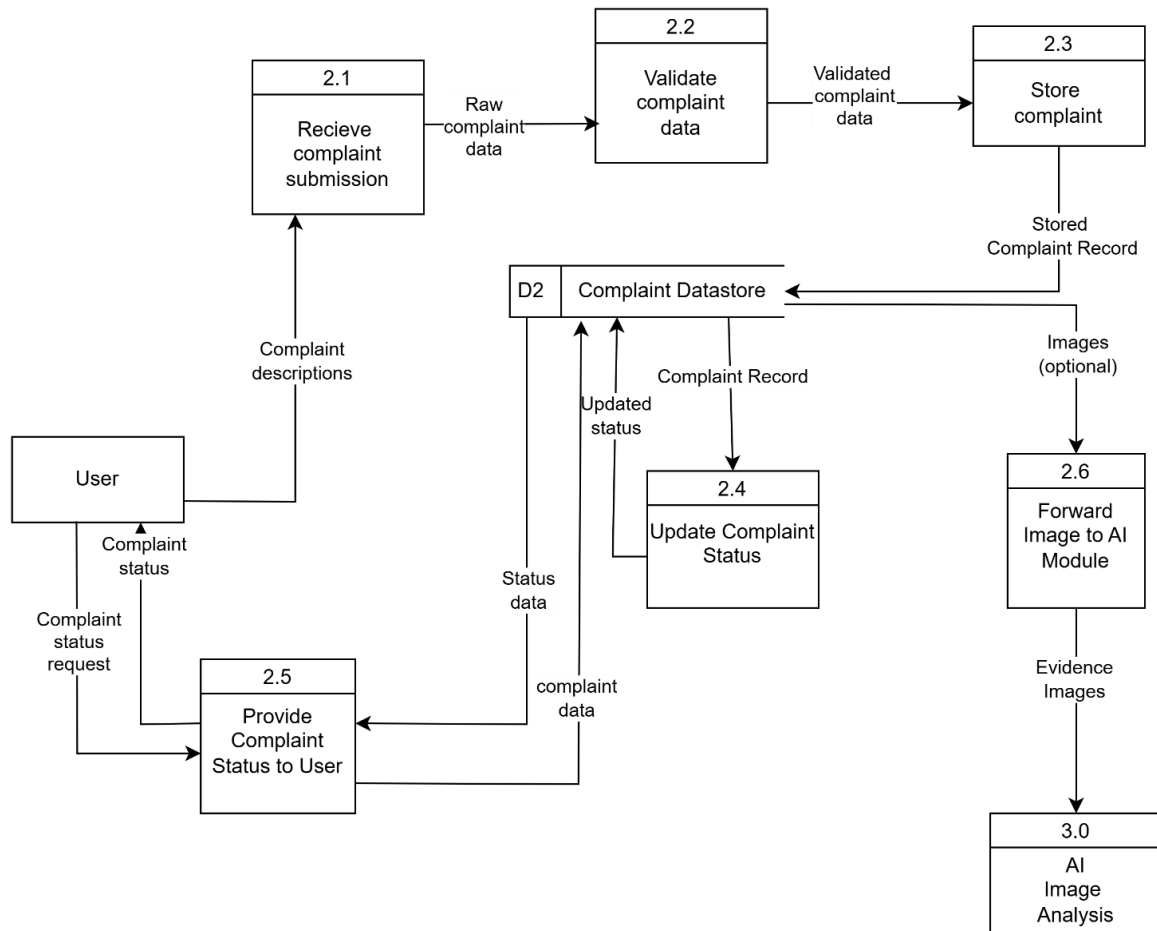


Figure 3.7: DFD level 2 for process 2.0

The Level 2 DFD of Complaint Management describes the workflow for receiving, validating, storing, and tracking complaints submitted by users.

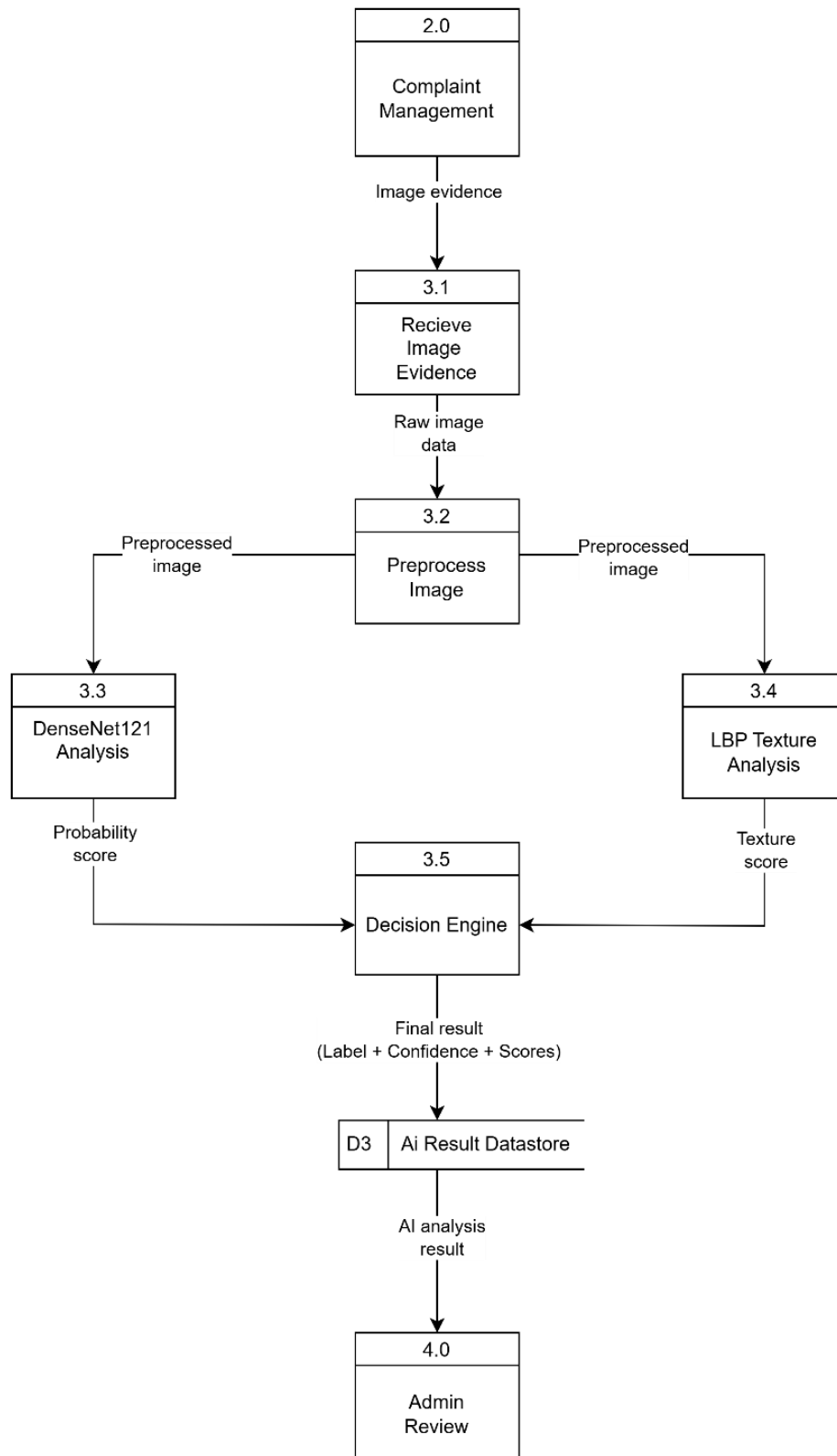


Figure 3.8: DFD level 2 for process 3.0

The Level 2 DFD of AI Image Analysis shows how uploaded image evidence is processed and analyzed using DenseNet121 and LBP techniques.

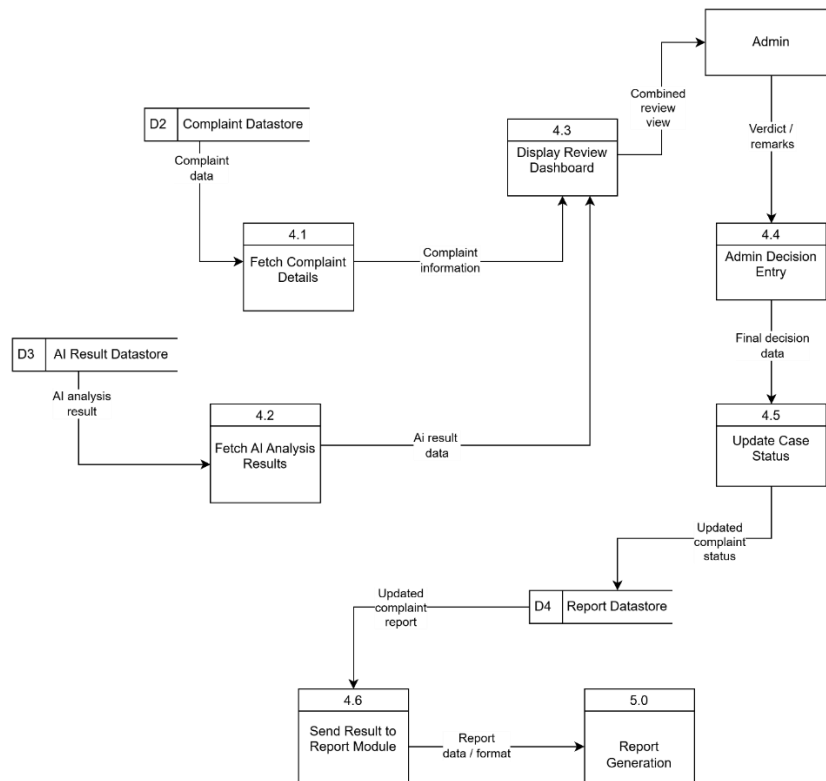


Figure 3.9: DFD level 2 for process 4.0

The Level 2 DFD of Admin Review illustrates how complaint information and AI analysis results are retrieved and presented to the administrator.

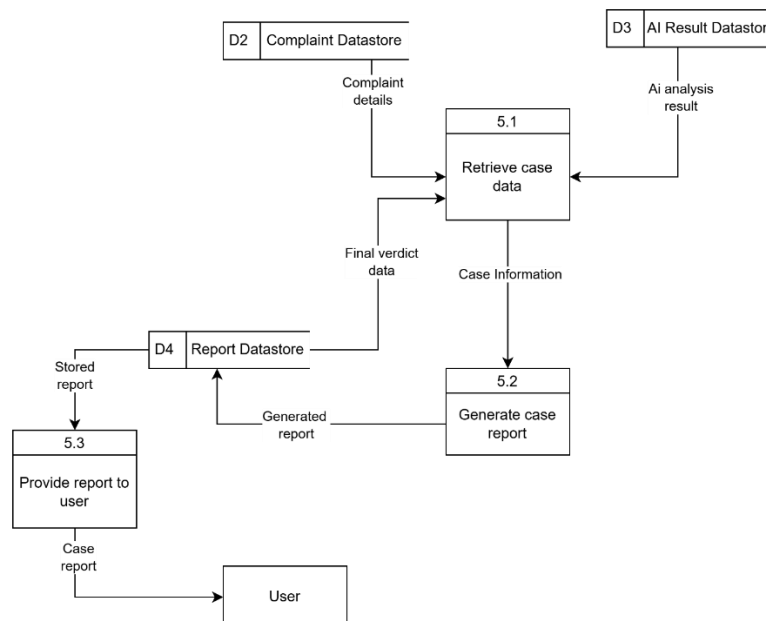


Figure 3.10: DFD level 2 for process 5.0

The Level 2 DFD of Report Generation depicts the process of collecting finalized case information and generating a report.

CHAPTER 4: SYSTEM DESIGN

System Design

This section describes the overall design of the Cybercrime Reporting System. The system is designed to integrate complaint management with AI-based image authenticity analysis in a modular and structured manner. The design follows a layered approach to ensure clarity, scalability, and maintainability.

4.1 Architectural Design

This section presents the overall architecture of the Cybercrime Reporting System. Users register, authenticate, and submit complaints with optional image evidence, which are stored in the PostgreSQL database. Complaint submission triggers the AI Detection module, where the image undergoes preprocessing, DenseNet121 classification, LBP feature extraction, and decision generation. The AI results are stored in the database and presented to administrators, who make the final review and decision.

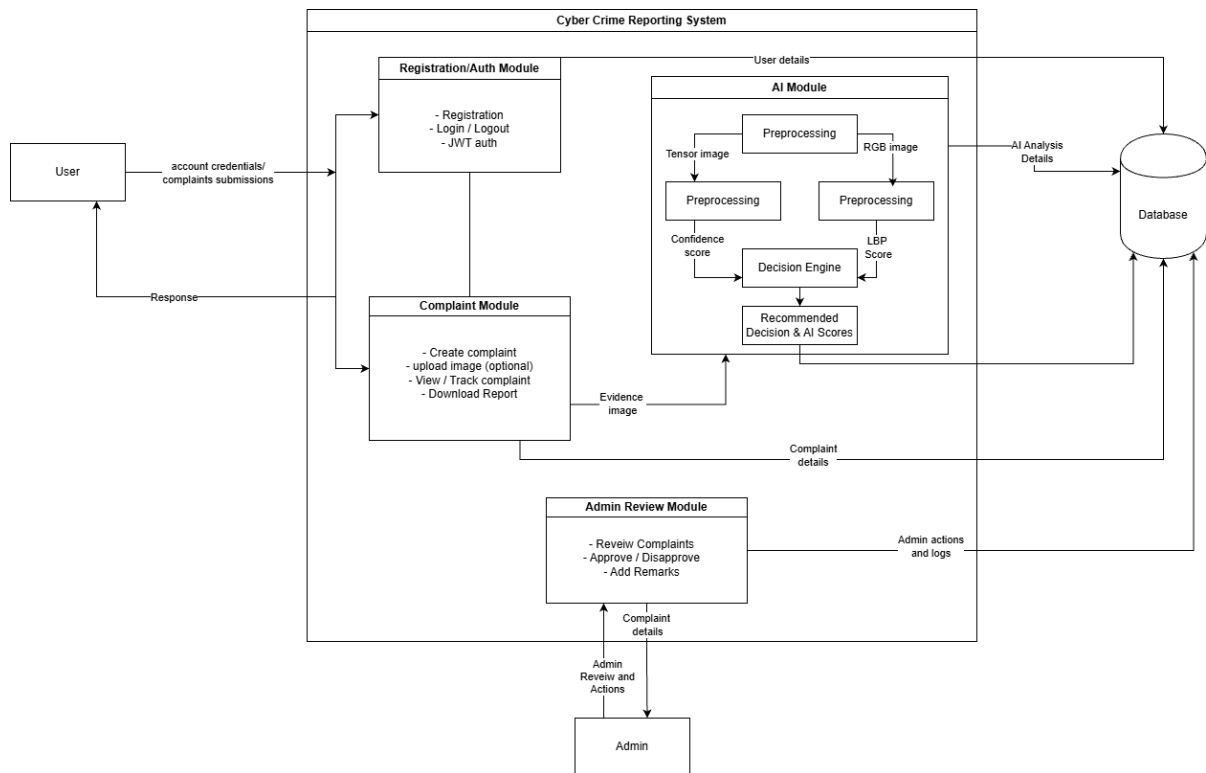


Figure 4.1 Architectural Design

4.2 Database Schema Design

The database is designed to manage user accounts, complaints, uploaded evidence of images, and AI analysis reports.

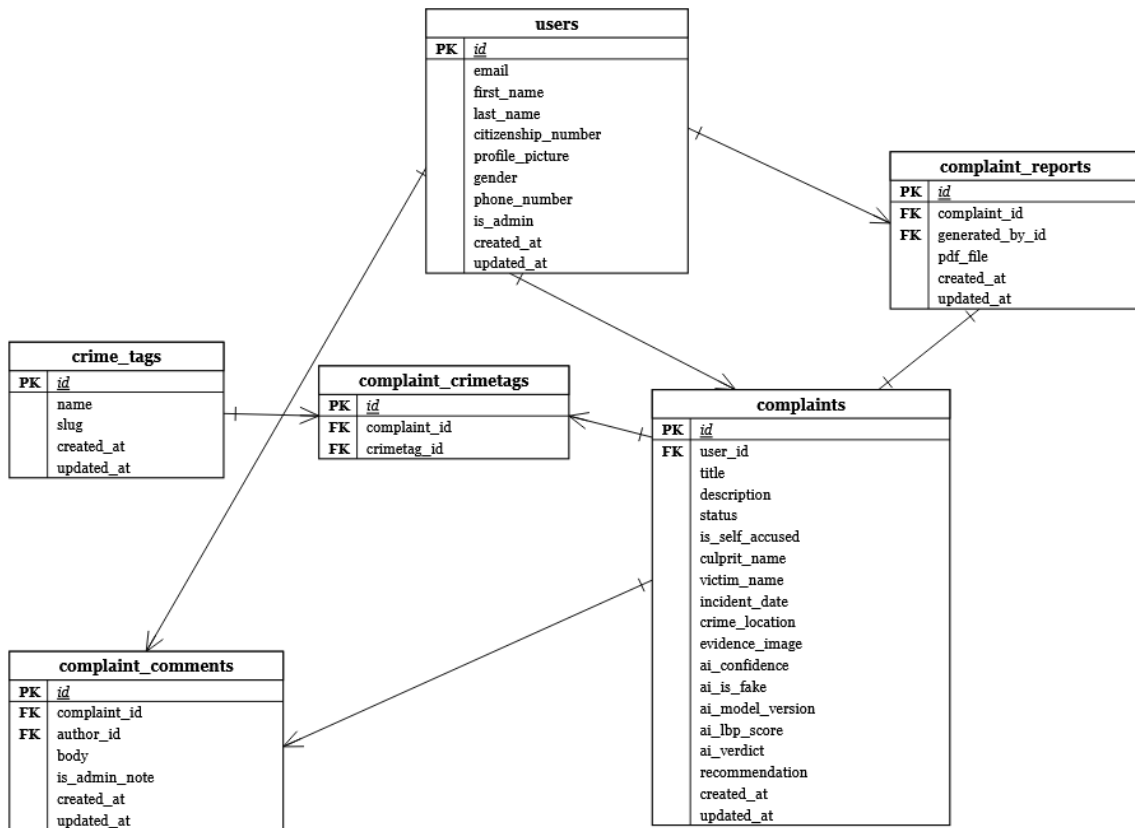


Figure 4.2: Database Schema Diagram

4.3 Process Design

The flowchart traces the full lifecycle of a system from initial registration and login through complaint submission, background AI analysis, admin review, and finally report generation and download.

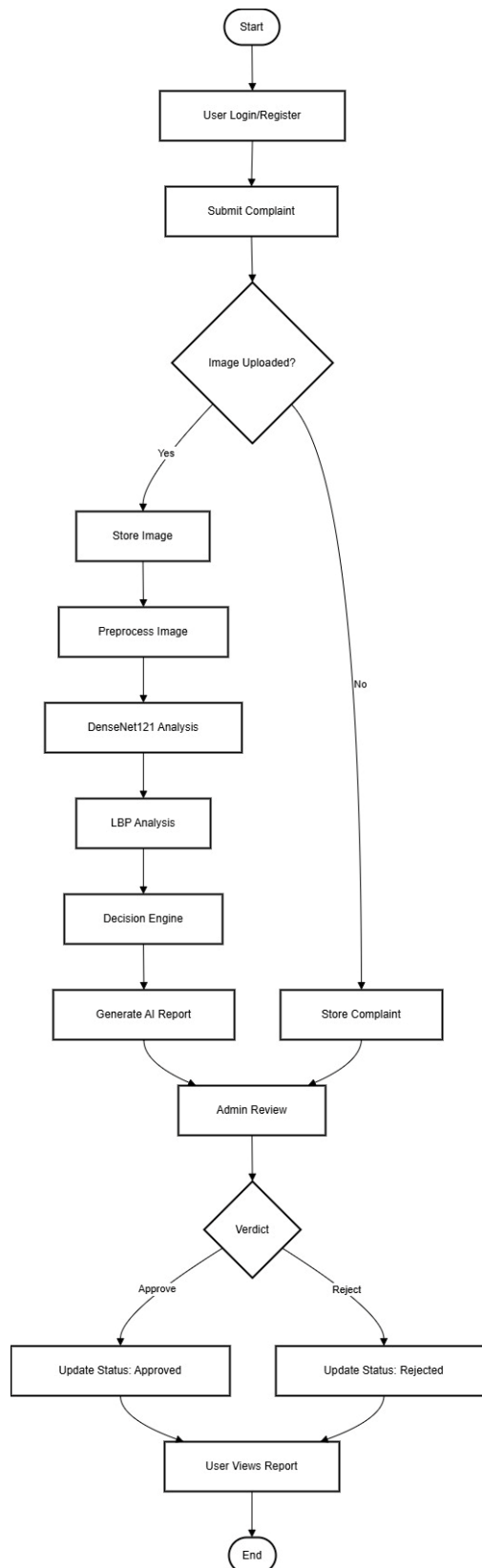


Figure 4.3: System Flowchart

4.4 Forms and Report Design

Forms and reports are the primary means through which users and administrators interact with the Cybercrime Reporting System. The login and registration forms are used to authenticate users and provide secure access to the system. The complaint submission form allows users to enter complaint details and upload supporting image evidence. The admin review form enables administrators to examine complaint information, review AI-generated analysis results, and update case statuses. The AI analysis report presents details of complaints, image authenticity analysis results, confidence scores, texture analysis findings, final classification labels, and administrator verdicts in a structured format.

Registration Form



The registration form is titled "Create an account" with the subtitle "Sign up to access the system". It is divided into three main sections: "PERSONAL INFORMATION", "CONTACT INFORMATION", and "ACCOUNT INFORMATION".

PERSONAL INFORMATION

- First Name:
- Middle Name:
- Last Name:
- Gender: ☐ Male ☐ Female ☐ Other
- Permanent Address:

CONTACT INFORMATION

- Email Address:
- Phone Number:

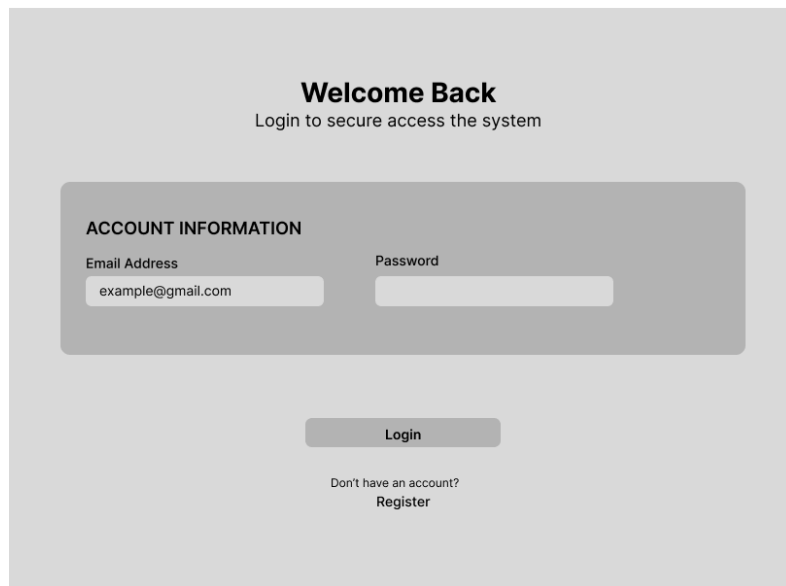
ACCOUNT INFORMATION

- Citizenship Number:
- Password:
- Confirm Password:

Already have an account?
[Login](#)

Figure 4.4: Registration Form Design

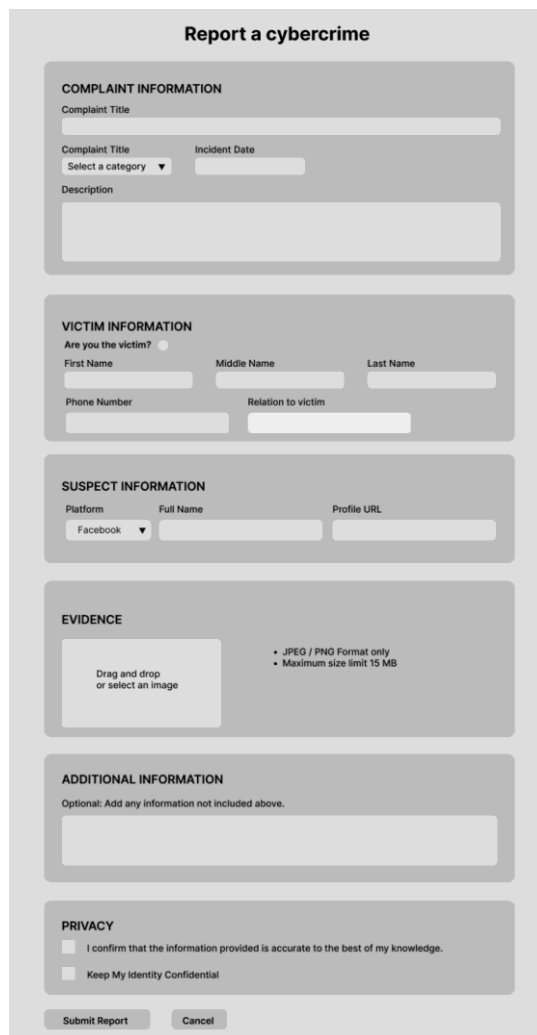
Login Form



The login form is centered on a light gray background. At the top, it says "Welcome Back" in bold, followed by "Login to secure access the system". Below this is a dark gray rounded rectangle containing the "ACCOUNT INFORMATION" section. This section has two input fields: "Email Address" with the placeholder "example@gmail.com" and "Password". Below the inputs is a "Login" button. At the bottom of the form, there is a link that says "Don't have an account? Register".

Figure 4.5: Login Form Design

Complaint Submission Form



The complaint submission form is titled "Report a cybercrime" and is divided into several sections. The "COMPLAINT INFORMATION" section includes a "Complaint Title" input field, a "Select a category" dropdown menu, an "Incident Date" input field, and a "Description" text area. The "VICTIM INFORMATION" section asks "Are you the victim?" with a radio button, followed by input fields for "First Name", "Middle Name", "Last Name", "Phone Number", and "Relation to victim". The "SUSPECT INFORMATION" section includes a "Platform" dropdown menu (set to "Facebook"), a "Full Name" input field, and a "Profile URL" input field. The "EVIDENCE" section features a "Drag and drop or select an image" area and a list of requirements: "JPEG / PNG Format only" and "Maximum size limit 15 MB". The "ADDITIONAL INFORMATION" section has a text area with the prompt "Optional: Add any information not included above.". The "PRIVACY" section contains two checkboxes: "I confirm that the information provided is accurate to the best of my knowledge." and "Keep My Identity Confidential". At the bottom are "Submit Report" and "Cancel" buttons.

Figure 4.6: Complaint Submission Form

Admin Review Form

Admin review form

COMPLAINT INFORMATION

Complaint Title

Online harassment

Category

Cyberbullying

Incident Date

2026-01-06

Description

Received hate speech and death threats in private message and social sites.

VICTIM INFORMATION

First Name

Rohan

Middle Name

Last Name

Kafie

Phone number

98xxxxxxxx

Relation to victim

Parent

SUSPECT INFORMATION

Platform

Facebook

Full Name

Profile URL

facebook.com/fake.binam.kaf

EVIDENCE

screenshot1.png

screenshot2.png

screenshot3.png

ADDITIONAL INFORMATION

REPORTER INFORMATION

First Name

Rohan

Middle Name

Last Name

Kafie

Phone Number

98xxxxxxxx

AI ANALYSIS REPORT

AI Verdict

Likely Manipulated

Confidence Score

92%

LBP Score

0.87

Recommended Decision

Further Review Recommended

REVIEW DECISION

Remarks

Approve

Disapprove

Figure 4.7: Admin Review Form

Cybercrime Complaint Review Report

CYBERCRIME COMPLAINT REVIEW REPORT

Report ID: CR-2026-00125
Complaint ID: CMP-2026-00087
Submission Date: 2026-01-06
Review Date: 2026-01-07

REPORTER INFORMATION

Reporter Name: Rohan Kafle
Email Address: rohankafle@example.com
Phone Number: 98XXXXXXX

COMPLAINT INFORMATION

Complaint Title:
Online harassment

Incident Date:
2026-01-06

Category:
Cyberbullying

Complaint Description:
Received hate speech and death threats in private message and social sites.

SUSPECT INFORMATION

Suspect Name:
Unknown

Platform:
Facebook

Profile URL:
facebook.com/fake.binam.kaf

EVIDENCE SUBMITTED

Evidence Image:


Additional Information:
None

AI ANALYSIS SUMMARY

Image Authenticity Result:
Likely Manipulated

Confidence Score:
92%

LBP Score:
0.87

REPORT STATUS

Status:
Pending

Remarks:
Evidence seems manipulated picture. Currently under review.

Generated By:
AI Based Cybercrime Reporting System

Figure 4.8: Cybercrime Complaint Review Report

4.5 Interface and Dialogue Design

The interface design follows a dual-role architecture, providing separate interfaces for complainants and administrators. User interfaces include registration, login, complaint submission, complaint tracking, and report viewing pages. Administrator interfaces include

complaint management, AI report review, and status update pages. All interfaces are designed to be simple, responsive, and user-friendly.

User Dashboard:

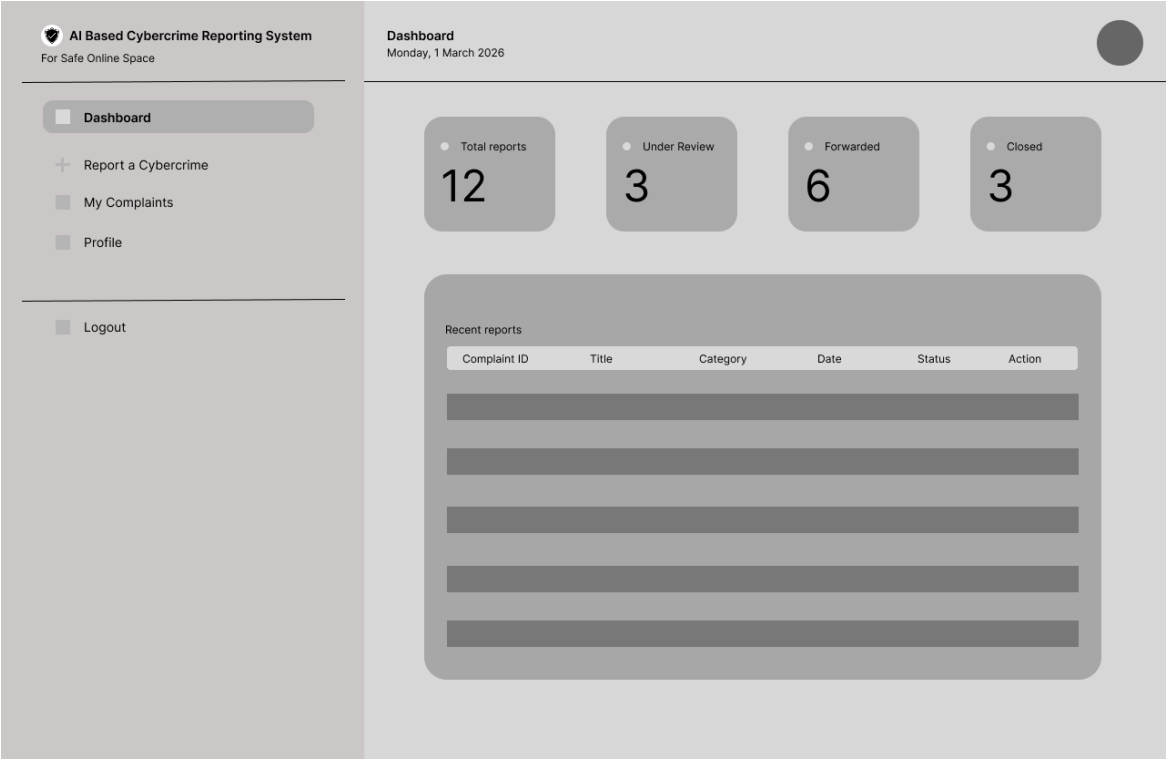


Figure 4.9: User Dashboard

Complaint Tracking Page:

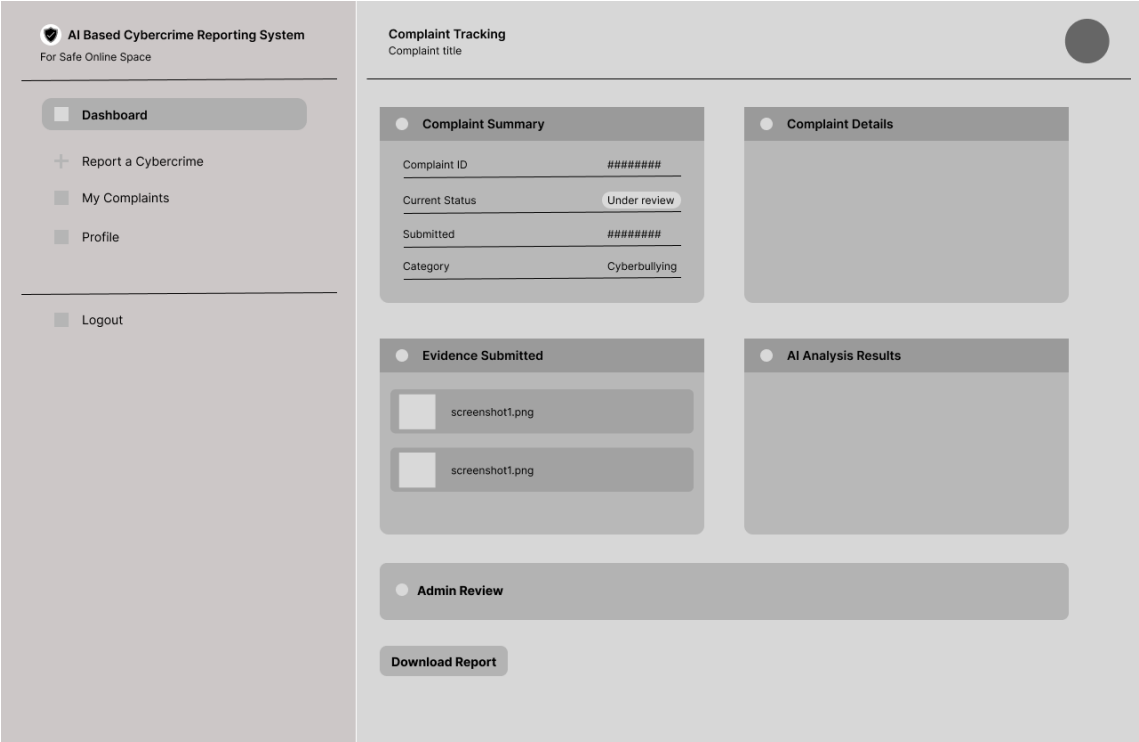


Figure 4.10: Complaint Tracking Page

Admin Dashboard:

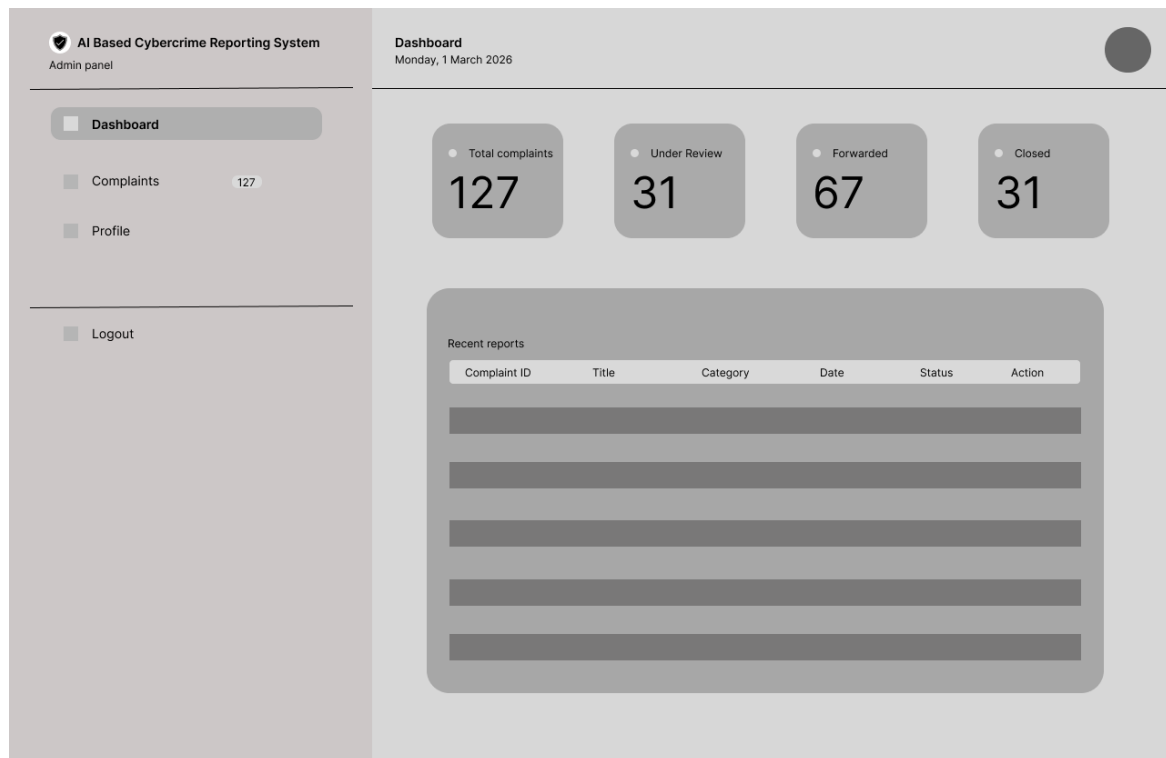


Figure 4.11: Admin Dashboard

Complaint Management Page:

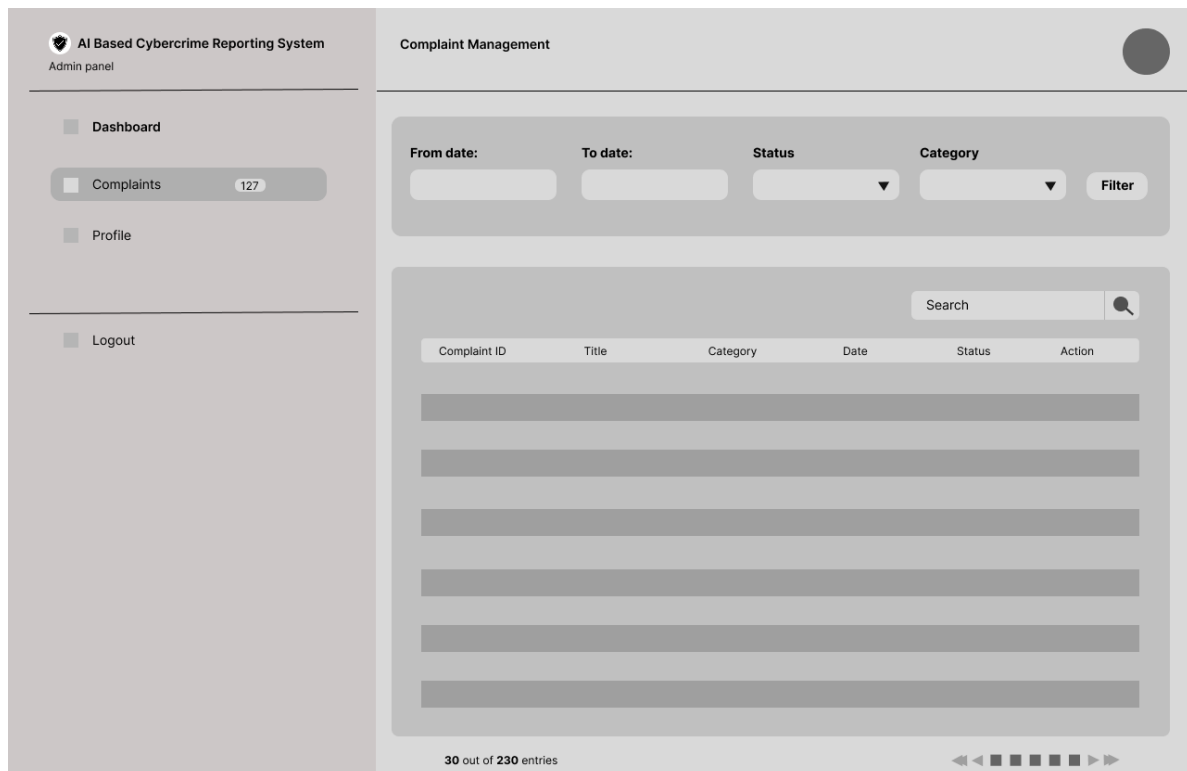


Figure 4.12: Complaint Management Page

4.6 Algorithm Details

4.6.1 DenseNet121

DenseNet121 (Densely Connected Convolutional Network) is the primary deep learning model used for image authenticity classification. Unlike conventional CNNs, each layer in DenseNet121 receives the feature maps of all preceding layers as input, promoting feature reuse, improving gradient propagation, and reducing the number of trainable parameters.

Mathematical Model,

For a DenseNet layer l , the output is computed as:

$$x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}]) \quad (4.1)$$

where:

x_l = output feature map of layer l

H_l = composite function of Batch Normalization, ReLU, and Convolution

$[x_0, x_1, \dots, x_{l-1}]$ = concatenation of feature maps from all preceding layers

The final prediction probability is obtained using the Softmax function:

where,

z_i = output logit of class

$C = 2$ (REAL, FAKE)

Table 4.1: Densenet121 Architecture

Layer	Configuration
Initial Convolution	7×7 Conv, 64 Filters
Dense Block 1	6 Layers
Transition Layer 1	1 × 1 Conv + Average Pooling
Dense Block 2	12 Layers
Transition 2	1 × 1 Conv + Average Pooling
Dense Block 3	24 Layers
Transition 3	1 × 1 Conv + Average Pooling
Dense Block 4	16 Layers
Global Average Pooling	1 Layer
Fully Connected Layer	Binary Classification

DenseNet121 Image Classification

Input: Image I

Output: Predicted class C and confidence score P

1. Read the input image I .
2. Resize the image to 64×64 pixels.
3. Normalize each pixel using ImageNet statistics:

$$x' = \frac{x - \mu}{\sigma} \quad (4.2)$$

where,

- $\mu = [0.485, 0.456, 0.406]$
 - $\sigma = [0.229, 0.224, 0.225]$
4. Convert the normalized image into tensor representation X .
 5. Pass X through the DenseNet121 network. Each dense block computes

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}]) \quad (4.3)$$

where

- $H_l(\cdot)$ denotes the composite function consisting of Batch Normalization, ReLU activation, and Convolution.
 - $[\cdot]$ represents feature-map concatenation.
6. Apply Global Average Pooling to obtain the feature vector:

$$g_k = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W f_k(i, j) \quad (4.4)$$

where,

- H and W are the height and width of the feature map.
 - $f_k(i, j)$ is the activation at location (i, j) .
7. Compute the output logits:

$$z = Wg + b \quad (4.5)$$

where,

- W is the weight matrix.
 - b is the bias vector.
8. Convert logits into class probabilities using the Softmax function:

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}} \quad (4.6)$$

Determine the predicted class:

$$C = \arg \max_i (P_i) \quad (4.7)$$

10. Return the predicted class C together with its confidence score P_i .

4.6.2 Local Binary Pattern (LBP)

Local Binary Pattern (LBP) is a texture descriptor used to capture local structural information within an image. The algorithm compares the intensity of each neighboring pixel with the center pixel and encodes the result into an 8-bit binary pattern.

Mathematical Model,

For a center pixel (x_c, y_c)

$$LBP(x_c, y_c) = \sum_{p=0}^{P-1} s(g_p - g_c) 2^p \quad (4.8)$$

where:

g_c = center pixel intensity

g_p = neighboring pixel intensity

P = number of neighboring pixels

The histogram of LBP values is

$$H(k) = \sum_{i=1}^N \delta(LBP_i, k) \quad (4.9)$$

where:

$H(k)$ = frequency of patterns

N = total number of pixels

δ = indicator function

The histogram is used as a texture feature vector.

Local Binary Pattern Extraction

Input: Image I

Output: Normalized LBP feature vector H_{norm}

1. Convert the input image I into grayscale.
2. For every non-border pixel, compute the Local Binary Pattern as

$$LBP(x_c, y_c) = \sum_{p=0}^7 s(g_p - g_c) \times 2^p \quad (4.10)$$

where,

- g_c is the gray value of the center pixel.
- g_p is the gray value of the p^{th} neighboring pixel.
- 3. The thresholding function is defined as

$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (4.11)$$

4. Repeat Step 2 until all pixels have been processed to generate the LBP image.
5. Construct the histogram of LBP values:

$$H(i) = \sum_x \sum_y f(LBP(x, y), i) \quad (4.12)$$

where,

$$f(a, b) = \begin{cases} 1, & a = b \\ 0, & a \neq b \end{cases} \quad (4.13)$$

6. Normalize the histogram:

$$H_{norm}(i) = \frac{H(i)}{\sum_{j=0}^{255} H(j)} \quad (4.14)$$

7. Return the normalized histogram H_{norm} as the LBP feature vector.

4.7 Evaluation Metrics

To evaluate the performance of the DenseNet121 model, several classification metrics were used. These metrics were calculated based on the values obtained from the confusion matrix, including True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

Accuracy

Accuracy measures the proportion of correctly classified images among all tested images.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (4.15)$$

Precision

Precision measures the proportion of images predicted as fake that are actually fake.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (4.16)$$

Recall

Recall measures the proportion of actual fake images that are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (4.17)$$

F1-Score

The F1-Score provides a balanced measure of precision and recall.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.18)$$

AUC-ROC

AUC-ROC measures the model's ability to distinguish between real and fake images across different classification thresholds. A value closer to 1.0 indicates better discriminative performance.

False Positive Rate (FPR)

The False Positive Rate measures the proportion of real images incorrectly classified as fake.

$$FPR = \frac{FP}{FP+TN} \quad (4.19)$$

False Negative Rate (FNR)

The False Negative Rate measures the proportion of fake images incorrectly classified as real.

$$FNR = \frac{FN}{FN+TP} \quad (4.20)$$

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1 Implementation

This phase focuses on developing the Cybercrime Reporting System by integrating backend services, AI modules, and APIs. The system processes complaints and performs image authenticity analysis when evidence is provided.

5.1.1 Tools Used

- **Python, Django (Backend Development):**

Django is used for authentication, complaint handling, database operations, and AI module integration, while Python supports overall system logic and machine learning implementation.

- **PyTorch (Deep Learning):**

PyTorch is used to implement the DenseNet121 model for image classification and AI-based image authenticity detection.

- **OpenCV, NumPy (Image Processing):**

OpenCV and NumPy are used for image preprocessing tasks such as resizing, grayscale conversion, normalization, and Local Binary Pattern (LBP) feature extraction.

- **PostgreSQL (Database):**

PostgreSQL is used as a database system to store user data, complaints, uploaded images, AI results, and generated reports.

- **ReactJS, HTML5, CSS3, Bootstrap 5 (Frontend Development):**

These technologies are used to build a responsive and interactive user interface for both users and administrators.

- **CIFAKE (Dataset):**

The CIFAKE dataset is used for training and testing the AI model, containing real and AI-generated images for classification tasks.

- **VS Code (IDE):**

Visual Studio Code is used as the development environment for coding, debugging, and project management.

- **Git/GitHub (Version Control):**

Git and GitHub are used for version control, code tracking, collaboration, and maintain project history.

5.1.2 Implementation Details of Modules

A. Authentication Module

The Authentication Module handles user registration, login, logout, and session management using JWT. A custom Django user model authenticates users with their email address, while access tokens are automatically refreshed to maintain active sessions.

```
const payload = JSON.parse(atob(stored.accessToken.split('.')[1]));
const isExpired = payload.exp * 1000 < Date.now();

if (isExpired && stored.refreshToken) {
  const { data } = await axios.post('/auth/refresh/', {
    refresh: stored.refreshToken
  });
  localStorage.setItem('accessToken', data.access);
}
```

Output

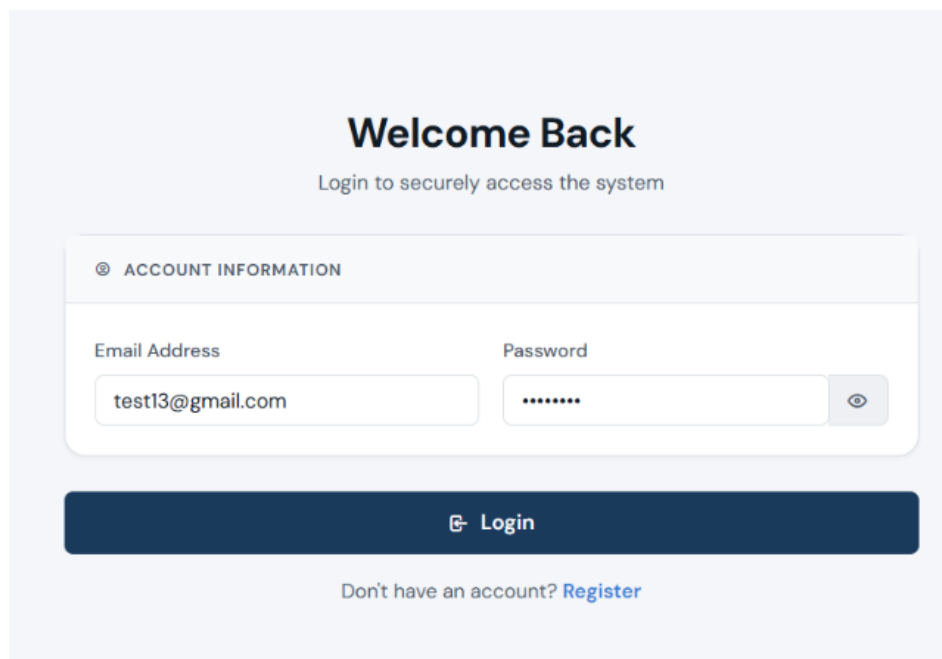


Fig 5.1: Authentication Module

B. Complaint Management Module

The complaint management module allows users to submit cybercrime complaints with optional image evidence. After successful submission, complaint information is stored in the database and forwarded to the AI analysis module when image evidence is present.

```
def create(self, validated_data):
    tags = validated_data.pop("tags", [])
    complaint = Complaint.objects.create(**validated_data)
```

```

complaint.tags.set(tags)

return complaint

def update(self, instance, validated_data):
    tags = validated_data.pop("tags", None)
    for attr, value in validated_data.items():
        setattr(instance, attr, value)
    instance.save()
    if tags is not None:
        instance.tags.set(tags)
    return instance

```

Output

Fig 5.2: Complaint submission form part 1

Fig 5.3: Complaint submission part 2

Fig 5.4: Complaint submission part 3

✔ Complaint submitted successfully

COMPLAINT ID	TITLE	CATEGORY	DATE	STATUS	ACTION
#12	Case of cyberbullying		05/07/2026	pending	<button>Track</button>

Fig 5.5: Complaint creation

C. Image Analysis Module (DenseNet121)

The DenseNet121 model is responsible for detecting whether the submitted image is likely authentic or AI-generated.

```
def load_model():
    global _model
    if _model is not None:
        return _model
    model = models.densenet121(weights=None)
    in_features = model.classifier.in_features
    model.classifier = torch.nn.Sequential(
        torch.nn.Dropout(0.3),
        torch.nn.Linear(in_features, 2)
    )
```

```

state_dict = torch.load(MODEL_PATH, map_location=DEVICE)

new_state = OrderedDict()
for k, v in state_dict.items():
    new_state[k.replace("module.", "")] = v

model.load_state_dict(new_state)
model.to(DEVICE).eval()

_model = model
return model

```

D. LBP Analysis Module

The LBP module provides texture-based analysis of the uploaded image. This module improves interpretability by identifying texture abnormalities commonly found in synthetic images.

```

def _compute_lbp_score(face_rgb) -> float:
    import numpy as np
    if face_rgb.ndim == 3:
        gray = (0.299 * face_rgb[:, :, 0] +
                0.587 * face_rgb[:, :, 1] +
                0.114 * face_rgb[:, :, 2]).astype(np.uint8)
    else:
        gray = face_rgb.astype(np.uint8)
    h, w = gray.shape
    lbp_map = np.zeros((h - 2, w - 2), dtype=np.uint8)
    offsets = [(-1, -1), (-1, 0), (-1, 1),
               (0, 1),
               (1, 1), (1, 0), (1, -1),
               (0, -1)]
    for i, (dy, dx) in enumerate(offsets):
        neighbour = gray[1+dy : h-1+dy, 1+dx : w-1+dx] if dy != 0 or dx != 0 \
            else gray[1:h-1, 1:w-1]
        center = gray[1:h-1, 1:w-1]

```

```

lbp_map |= ((neighbour >= center).astype(np.uint8) << i)
hist, _ = np.histogram(lbp_map.ravel(), bins=256, range=(0, 256), density=True)
score = float(hist.max())
return round(min(score * 10, 1.0), 4)

```

E. Decision Engine Module

The Decision Engine combines DenseNet121 results with LBP outputs.

Table 5.1: Decision Rules

AI Score	LBP Result	Final Label
> 0.70	Irregular	Manipulated
0.40–0.70	Mixed	Suspicious
< 0.40	Normal	Likely Real

```

def _derive_verdict_and_decision(is_fake: bool, confidence_pct: float, lbp_score: float):
    from .models import AIVerdict, RecommendedDecision
    if not is_fake and confidence_pct < 60:
        verdict = AIVerdict.INCONCLUSIVE
    elif not is_fake:
        verdict = AIVerdict.AUTHENTIC
    elif confidence_pct >= 85 or lbp_score >= 0.80:
        verdict = AIVerdict.FAKE
    else:
        verdict = AIVerdict.LIKELY_MANIPULATED
    if verdict == AIVerdict.AUTHENTIC:
        decision = RecommendedDecision.NO_ACTION
    elif verdict == AIVerdict.INCONCLUSIVE:
        decision = RecommendedDecision.FURTHER_REVIEW
    elif verdict == AIVerdict.LIKELY_MANIPULATED:
        decision = RecommendedDecision.ESCALATE
    else:
        decision = RecommendedDecision.REFER_FOR_PROSECUTION
    return verdict, decision

```

Output

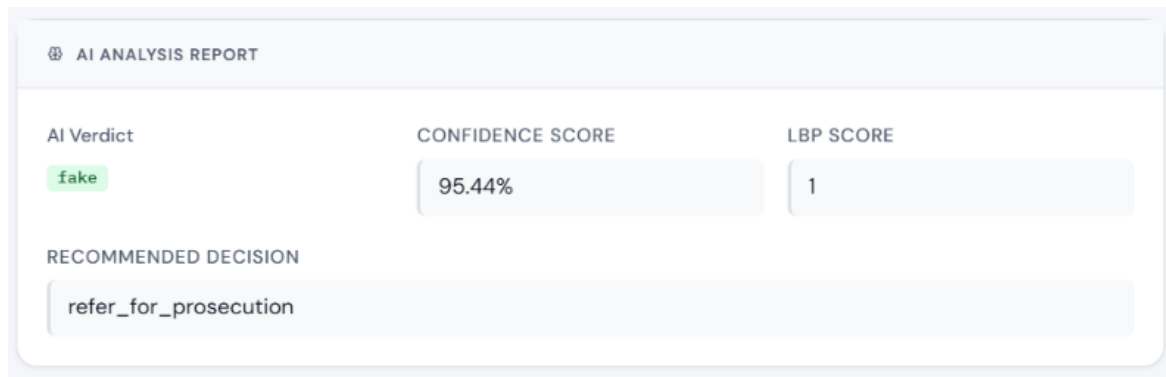


Fig 5.6: AI Analysis Output from Decision Engine

F. Report Generation Module

The report generation module creates an AI-assisted analysis report after image evaluation.

The report is stored in the database and made available to both administrators and users.

```
def post(self, request, pk):
    complaint = get_object_or_404(Complaint, pk=pk)
    report, _ = ComplaintReport.objects.get_or_create(
        complaint=complaint,
        defaults={"generated_by": request.user},
    )
    try:
        report.generate_pdf()
    except Exception as e:
        return Response(
            {"error": f"PDF generation failed: {str(e)}"},
            status=status.HTTP_500_INTERNAL_SERVER_ERROR,
        )
    serializer = ComplaintReportSerializer(report, context={"request": request})
    return Response(serializer.data, status=status.HTTP_200_OK)
```

Output

Crime Complaint Report

Case #1 | Generated at: 26 Jun 2026, 08:57

Case Overview

Status	Approved
Filed	26 Jun 2026, 05:43
Last Updated	26 Jun 2026, 09:03

Complainant Information

Identity	Anonymous Submission
----------	----------------------

Incident Details

Title	TEST ONE
Incident Date	08 Jun 2026
Crime Location / Platform	facebook
Alleged Perpetrator	Dan Issac
Victim Name	Don Lissac
Victim Phone	68787
Self-Accused	No

Complaint Description

Where does it come from? Contrary to popular belief, Lorem Ipsum is not simply random text. It has roots in a piece of classical Latin literature from 45 BC, making it over 2000 years old. Richard McClintock, a Latin professor at Hampden-Sydney College in Virginia, looked up one of the more obscure Latin words, consectetur, from a Lorem Ipsum passage, and going through the cites of the word in classical literature, discovered the undoubtable source. Lorem Ipsum comes from sections 1.10.32 and 1.10.33 of "de Finibus Bonorum et Malorum" (The Extremes of Good and Evil) by Cicero, written in 45 BC. This book is a treatise on the theory of ethics, very popular during the Renaissance. The first line of Lorem Ipsum, "Lorem ipsum dolor sit amet..", comes from a line in section 1.10.32. The standard chunk of Lorem Ipsum used since 1966 is reproduced below for those interested. Sections 1.10.32 and 1.10.33 from "de Finibus Bonorum et Malorum" by Cicero are also reproduced in their exact original form, accompanied by English versions from the 1914 translation by H. Rackham.

Fig 5.7: Report Generation Format Part 1

AI Image Analysis

AI Verdict	Authentic
Confidence Score	97.6%
LBP Score	1.00
Recommended Decision	No Action Required
Model Version	densenet121_v1
AI Flagged	No



Submitted evidence

This report is confidential and intended solely for authorised investigative use. Unauthorised disclosure is prohibited.
Generated by Crime Portal v1.0.

Fig 5.8: Report Generation Format Part 2

5.2 Testing

Testing was conducted to verify that all modules operate correctly and that the system performs as expected under normal conditions.

Testing was divided into:

1. Unit Testing
2. System Testing

5.2.1 Unit Testing

Unit testing was performed on individual modules independently.

1. Authentication Module

Table 5.2: Authentication Testing

S.N.	Test Input	Expected Output	Actual Output	Result
1	Register with valid name, unique email, strong password and valid fields	Account is created successfully	Account created successfully	Pass
2	Register using an existing email	Registration fails with "Email already exists."	"Email already exists." displayed	Pass
3	Register with weak password (12345)	Password validation error	Password validation error displayed	Pass
4	Login with correct email and password	User is redirected to dashboard	User redirected to dashboard	Pass
5	Login with incorrect password	Login fails with invalid credentials message	Invalid credentials message displayed	Pass
6	Login with empty email and password	Validation errors shown	Validation errors shown	Pass

Create an Account

Sign up to access the system

PERSONAL INFORMATION

First Name *

Middle Name

Last Name *

Test

user

Gender *

☐ Male
☒ Female
☐ Other

Permanent Address *

Kamalvinayak, Bhaktapur

CONTACT INFORMATION

Email Address *

Phone Number *

test1@gmail.com

9812345678

ACCOUNT INFORMATION

Citizenship Number *

bc-de-fg-hijkl

Password *

Confirm Password *

Register

Already have an account? Login

Fig 5.9: Registration

Create an Account

Sign up to access the system

PERSONAL INFORMATION

First Name *

Middle Name

Last Name *

This field is required.

This field is required.

Gender *

☐ Male
☐ Female
☐ Other

This field is required.

Permanent Address *

This field is required.

CONTACT INFORMATION

Email Address *

Phone Number *

example@gmail.com

98XXXXXXX

This field is required.

This field is required.

ACCOUNT INFORMATION

Citizenship Number *

XX-XX-XX-XXXX

This field is required.

Password *

Confirm Password *

This field is required.

This field is required.

Register

Already have an account? Login

Fig 5.10: Failed Registration

Create an Account

Sign up to access the system

PERSONAL INFORMATION

First Name *

Middle Name

Last Name *

Gender *

☐ Male
 ☒ Female
 ☐ Other

Permanent Address *

CONTACT INFORMATION

Email Address *

Phone Number *

ACCOUNT INFORMATION

Citizenship Number *

Password *

Confirm Password *

Register

Already have an account? [Login](#)

1

user with this email already exists., user with this citizenship number already exists.

Fig 5.11: Duplicate account registration failed

Welcome Back

Login to securely access the system

ACCOUNT INFORMATION

Email Address

Password

Login

Don't have an account? [Register](#)

Fig 5.12: Login

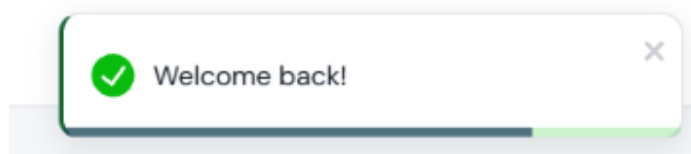


Fig 5.13: Successful Login popup

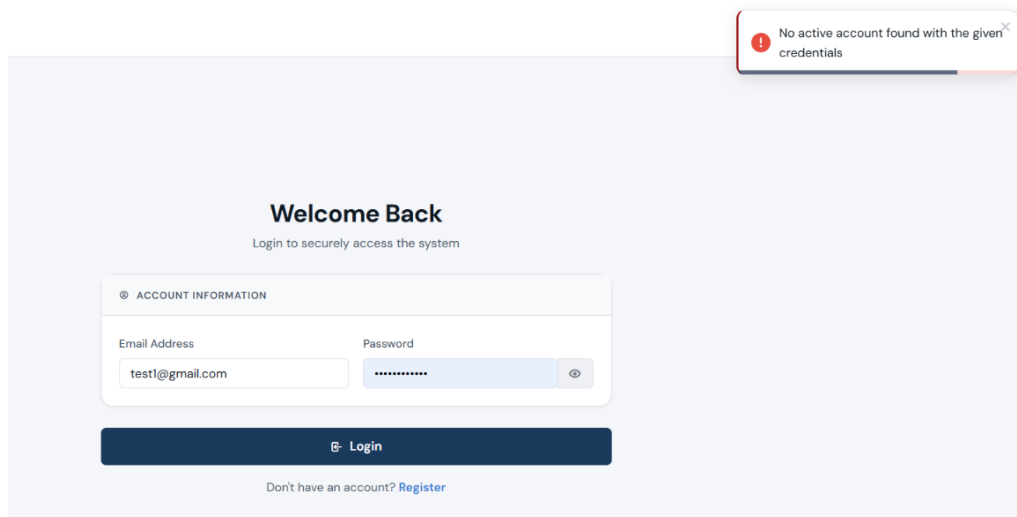


Fig 5.14: Failed Login

2. Complaint Module

Table 5.3: Complaint Submission Testing

S.N.	Test Input	Expected Output	Actual Output	Result
1	Submit complaint with all required fields and valid image	Complaint stored successfully	Complaint stored successfully	Pass
2	Submit complaint without an image	Complaint stored successfully	Complaint stored successfully	Pass
3	Submit complaint with a valid JPEG image	Complaint submitted and JPEG image accepted	Complaint submitted and JPEG image accepted	Pass
4	Submit complaint with a valid PNG image	Complaint submitted and PNG image accepted	Complaint submitted and PNG image accepted	Pass
5	Submit complaint with a GIF image	GIF image rejected with validation message	GIF image rejected with validation message	Pass

6	Submit complaint with an unsupported file type (e.g., PDF)	File upload rejected	File upload rejected	Pass
7	Submit complaint with an image exceeding the maximum size (e.g. 19MB)	Upload rejected with validation message	Upload rejected with validation message	Pass
8	Submit complaint with empty required fields	Validation errors displayed; complaint not submitted	Validation errors displayed	Pass
9	Submit complaint while not logged in	Access denied or redirected to login page	Redirected to login page	Pass
10	Review complaint from Admin Page and approve with a remarks	Complaint approved successfully with remarks	Complaint approved but remarks not shown	Fail

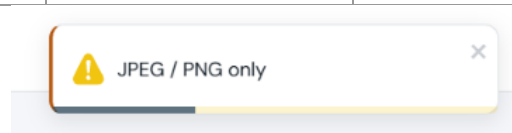


Fig 5.15: Form Submission Error Popup when input is GIF

Fig 5.16: Form submission without image upload part 1

VICTIM INFORMATION

☐ I am the victim

First Name *
testfname

Middle Name
testmname

Last Name *
testlname

Phone Number *
9812345678

Relation to Victim *
friend

SUSPECT INFORMATION

Platform
Facebook

Full Name / Username
dotdot

Profile URL
https://...

Fig 5.17: Form submission without image upload part 2

EVIDENCE



Drag & drop or click to upload

JPEG / PNG - Max 15 MB

- JPEG / PNG format only
- Maximum size 15 MB
- One image — screenshot or photo of incident
- Image will be analysed by AI for manipulation

PRIVACY

☒ I confirm that the information provided is accurate to the best of my knowledge. *

☐ Keep my identity confidential

Submit Report
Cancel

Fig 5.18: Form submission without image upload part 3

Complaint submitted successfully

COMPLAINT ID	TITLE	CATEGORY	DATE	STATUS	ACTION
#13	test title		06/07/2026	pending	Track

Fig 5.19: Success form submission without image upload

3. AI Detection Module

Table 5.4: AI Detection Module Testing

S.N.	Test Input	Expected Output	Actual Output	Result
1	Upload authentic image	Classified as Likely Real	Classified as Likely Real	Pass
2	Upload AI-generated image	Classified as AI Generated	Classified as likely manipulated	Pass
3	Upload low-quality image	Analysis completed	Classified as Inconclusive	Pass
4	Upload manipulated image	Suspicious result generated	Classified as likely manipulated	Pass
5	Upload corrupted image	Recommended for rejected	Classified as inconclusive	Fail

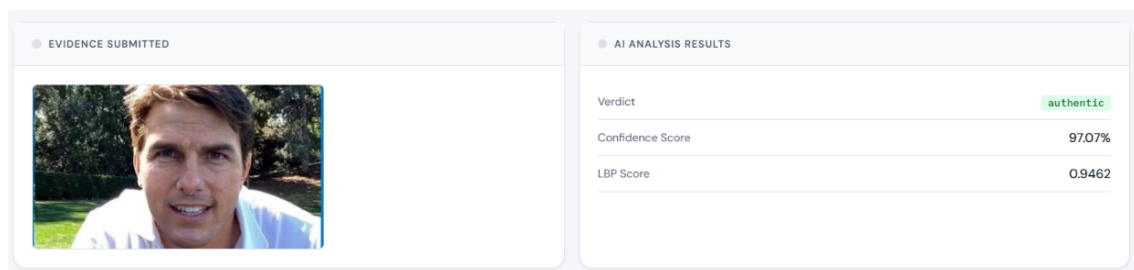


Fig 5.20: Testing with real picture

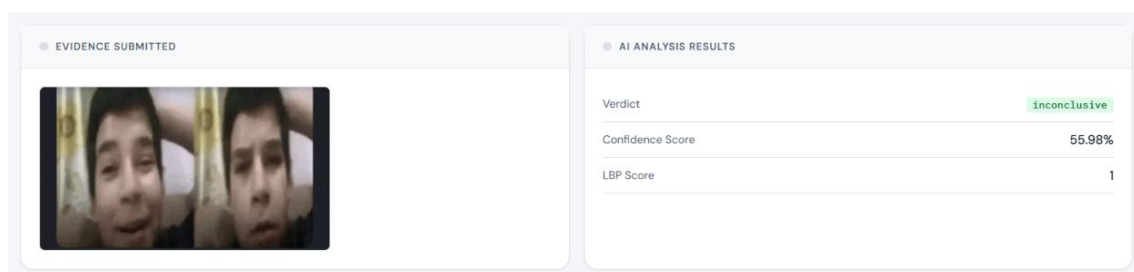


Fig 5.21: Testing with low quality and mixed textures

5.2.2 System Testing

System testing evaluated the complete workflow of the application.

Table 5.5: System Testing

S.N.	Test Scenario	Expected Output	Actual Output	Result
1	User registers with valid credentials	Account created successfully	Account created successfully	Pass
2	User logs in with valid credentials	Access granted to dashboard	Access granted to dashboard	Pass
3	User submits a complaint with a valid image	Complaint stored successfully	Complaint stored successfully	Pass
5	AI detection process is executed	Detection result generated successfully	Detection result generated successfully	Pass
6	Admin reviews and updates complaint status	Complaint status updated	Complaint status updated	Pass
7	User downloads the analysis report	Report downloaded successfully	Report downloaded successfully	Pass

The complete workflow executed successfully without system failures.

5.3 Result Analysis

The performance of the DenseNet121 model was evaluated using the CIFAKE test dataset, which consisted of 20,000 unseen images that were not used during the training phase. The evaluation was conducted to measure the model's ability to accurately distinguish between authentic and AI-generated images.

Table 5.6: Training Configuration

Parameter	Value
Architecture	DenseNet121
Input Size	$64 \times 64 \times 3$
Classes	REAL, FAKE
Optimizer	Adam

Learning Rate	0.0003
Batch Size	128
Epochs	10
Loss Function	Cross Entropy Loss

Table 5.7: Model Performance

Metric	Value
Accuracy	98.61%
Precision	98.78%
Recall	98.45%
F1-Score	98.61%
AUC-ROC	99.79%
False Positive Rate	1.22%
False Negative Rate	1.55%

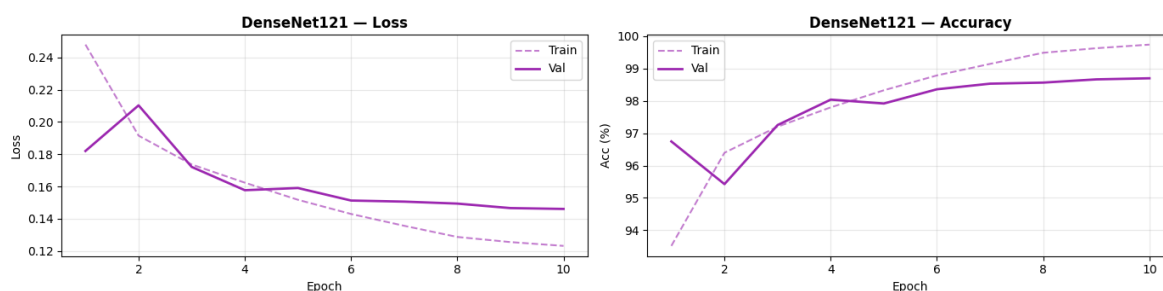


Figure 5.22: Training and Validation Curve

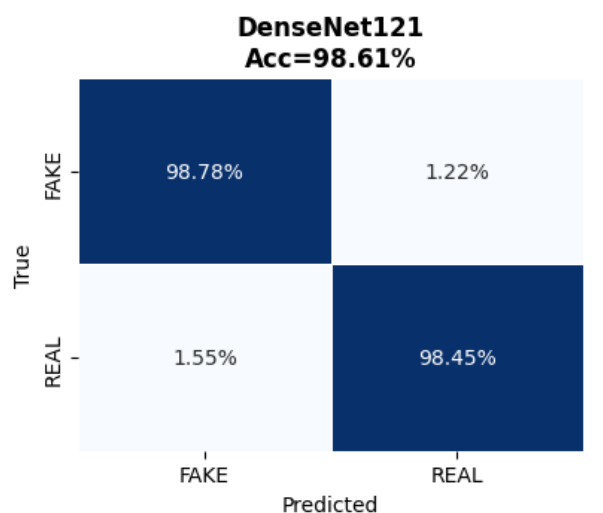


Figure 5.23: Confusion Matrix

Training and validation analysis

The training and validation curves shown in Figure 5.22 demonstrate that the DenseNet121 model achieved stable learning throughout the training process. The training and validation accuracy increased consistently over the epochs, while the loss values decreased gradually. This indicates that the model was able to learn meaningful image features without significant overfitting.

Confusion matrix analysis

The confusion matrix presented in Figure 5.23 illustrates the distribution of correctly and incorrectly classified images. Out of the 20,000 test images, the model correctly classified 19,722 images, resulting in an overall accuracy of 98.61%. The small number of false positives and false negatives indicates that the model has a strong capability to distinguish between real and AI-generated images.

Performance interpretation

The precision of 98.78% indicates that when the model predicts an image as manipulated, it is correct 98.78% of the time. This is important in cybercrime investigations because it reduces the likelihood of falsely identifying authentic evidence as manipulated.

The recall of 98.45% reflects the model's ability to correctly identify actual manipulated images. A high recall value is desirable because it minimizes the number of manipulated images that may be incorrectly classified as authentic.

The F1-score of 98.61% confirms a strong and balanced trade-off between precision and recall, indicating that the model performs consistently across both classes without disproportionately favoring one class over the other.

The AUC-ROC value of 99.79% demonstrates excellent discriminative capability. A value close to 1.0 indicates that the model can effectively separate authentic images from AI-generated images across different classification thresholds, providing confidence in the reliability of the generated AI score used within the decision engine.

Overall analysis

The experimental results demonstrate that the proposed DenseNet121-based image authenticity classification model performs effectively on the CIFAKE dataset. The combination of DenseNet121 for deep feature extraction and Local Binary Pattern (LBP) for texture analysis provides a reliable AI-assisted mechanism for detecting manipulated or AI-generated images. The high accuracy, precision, recall, F1-score, and AUC-ROC values indicate that the model is suitable for integration into the Cybercrime Reporting System to support administrative review and digital evidence assessment.

CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

The Cybercrime Reporting System is developed as a web-based platform to address the growing challenges of cybercrime involving manipulated and AI-generated images. By integrating AI-assisted image analysis into a structured complaint workflow, the system improves the efficiency and consistency of digital evidence evaluation at the initial reporting stage.

The system utilizes DenseNet121 for image authenticity analysis and Local Binary Pattern (LBP) for supporting texture-based insights. These components work together through a rule-based decision engine to generate interpretable AI-assisted reports. This approach provides a more structured and technically supported method compared to traditional systems that rely solely on manual inspection.

The platform offers a user-friendly interface for complainants to submit reports and enables administrators to review complaints along with AI-generated analysis. The human-in-the-loop design ensures that final decisions remain under administrative control, maintaining accountability and reliability.

From a technical perspective, the system is implemented using Django and PyTorch, ensuring modularity, scalability, and efficient performance. The development process provided practical experience in integrating AI models into real-world applications, designing system architecture, and managing end-to-end workflows.

Overall, the project demonstrates that AI-assisted cybercrime reporting systems can enhance the reliability, speed, and consistency of complaint processing while supporting human decision-making.

6.2 Future Recommendations

While the current system provides a strong and functional foundation for AI-assisted cybercrime complaint management, several enhancements are proposed for future development:

1. **Video Analysis Support** The current system processes only static image evidence, limiting its applicability to cases involving video-based submissions, which are increasingly common in cybercrime complaints. Future versions of the system should

incorporate video analysis capabilities using frame-level extraction and temporal consistency analysis, enabling the detection of manipulated or deepfake video content.

2. **Higher-Resolution Model Training** The DenseNet121 model was trained on the CIFAKE dataset, which uses 64×64 pixel images. Real-world submissions are likely to be significantly higher in resolution. Future work should explore training or fine-tuning the model on higher-resolution datasets such as FaceForensics++ [5] to improve detection accuracy across a broader range of image qualities.
3. **Multi-Image Support** The current implementation accepts only one evidence image per complaint. In practice, complainants may need to submit multiple images to fully document an incident. A future version should support multi-image uploads, with individual analysis results combined to produce an aggregated assessment at the complaint level.
4. **Metadata and EXIF Forensic Analysis** Digital images often contain embedded metadata, including EXIF data, recording information such as the camera make and model, GPS coordinates, and timestamps. Incorporating automated EXIF analysis could provide an additional layer of evidence validation, particularly for identifying inconsistencies between image content and its reported origin.
5. **User Feedback and Continuous Learning** Introducing a structured feedback mechanism would allow administrators to flag incorrect AI classifications, providing labelled data that can be used to periodically retrain and improve the model. This feedback loop would allow the system to adapt over time to emerging types of AI-generated content and manipulation techniques.
6. **Integration with External Law Enforcement Systems** The current system operates as a standalone platform. Future development should explore integration with national or institutional law enforcement case management systems, enabling complaint data and AI analysis reports to be shared securely across agencies and improving coordination in cybercrime investigations.

REFERENCES

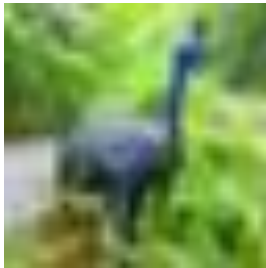
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APPENDICES

A1 Fake Image from Dataset



A2 Real image from Dataset



B1

← → ↺ 🌐 localhost:5173/profile

**AI Based
Cybercrime Reporting
System**
For Safe Online Space

 Dashboard

 + Report a Cybercrime

 My Complaints

 **Profile**

Profile

Manage your account information

**Bikram Gharti**
test13@gmail.com
USER

PERSONAL INFORMATION

First Name

Bikram

Middle Name

Last Name

Gharti

Phone Number

Permanent Address

Citizenship Number

ab-cd-ef-ghijk

Email Address

test13@gmail.com



CHANGE PASSWORD

Current Password

**Bikram Gharti**
test13@gmail.com

 Sign out

B2

AI Based Cybercrime Reporting System
For Safe Online Space

Dashboard

+ Report a Cybercrime

My Complaints

Profile

Bikram Gharti
test13@gmail.com

Sign out

System operational

My Complaints
All complaints you have submitted

From date
mm/dd/yyyy

To date
mm/dd/yyyy

Status
All

Category
All

Filter

X

Search...

COMPLAINT ID	TITLE	CATEGORY	DATE	STATUS	ACTION
#13	test title		06/07/2026	pending	Track
#12	Case of cyberbullying		05/07/2026	pending	Track
#11	csrf crime 1		05/07/2026	ongoing	Track
#9	test crime 1		05/07/2026	ongoing	Track
#8	testinggg		12/06/2026	ongoing	Track

B3

AI Based Cybercrime Reporting System
For Safe Online Space

Dashboard

+ Report a Cybercrime

My Complaints

Profile

Bikram Gharti
test13@gmail.com

Sign out

System operational

SUSPECT INFORMATION

Platform
Facebook

Full Name / Username
Known name or username

Profile URL
https://...

EVIDENCE

Drag & drop or click to upload
JPEG / PNG - Max 15 MB

- JPEG / PNG format only
- Maximum size 15 MB
- One image – screenshot or photo of incident
- Image will be analysed by AI for manipulation

PRIVACY

☐ I confirm that the information provided is accurate to the best of my knowledge. *

☐ Keep my identity confidential

Submit Report

Cancel

B4

AI Based Cybercrime Reporting System
For Safe Online Space

Dashboard

+ Report a Cybercrime

My Complaints

Profile

Bikram Gharti
test13@gmail.com

Sign out

System operational

Complaint Tracking
testing 3

← Back

Download Report

PENDING

ONGOING

RESOLVED

COMPLAINT SUMMARY

COMPLAINT DETAILS

EVIDENCE SUBMITTED

AI ANALYSIS RESULTS

Complaint ID
#3

Current Status
ongoing

Submitted
Jun 7, 2026

Category
—

Title
testing 3

Incident Date
—

Platform
Facebook —

Description
—



Verdict
authentic

Confidence Score
83.72%

LBP Score
1

B5

AI Based
Cybercrime Reporting
System
Admin Panel

Dashboard

Complaints

Profile

Admin session

approved + Back

Admin Review Form

Complaint #19

COMPLAINT INFORMATION

COMPLAINT TITLE

kjhkhkhkhk

CATEGORY

INCIDENT DATE

Financial Fraud

05/07/2026

Description

xx

VICTIM INFORMATION

FIRST NAME

MIDDLE NAME

LAST NAME

hbjhbjh

kjkj

kjnklnk

PHONE NUMBER

RELATION TO VICTIM

B6

PLATFORM	FULL NAME / USERNAME	PROFILE URL
Facebook —	—	—

 EVIDENCE



 ADDITIONAL INFORMATION

B7

AI ANALYSIS REPORT

AI Verdict

CONFIDENCE SCORE

LBP SCORE

fake

50.34%

1

RECOMMENDED DECISION

refer_for_prosecution

REVIEW DECISION

Remarks *

Enter your review remarks...

Approve

Disapprove

Log of Visits to Supervisor

Visit No.	Date	Technical Feedback	Supervisor's Signature
1	2082/12/18	Initial Discussion about Project.	
2	2082/12/25	Proposal for the project submitted.	
3	2083/02/05	Discussion about report writing	
4	2083/04/01	Project Demo and Report Submission	
5	2083/04/08	Report Resubmission and Discussion	